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Designing for Consent Revocation in Online Platforms

Final Project Report

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Abstract

Digital messaging platforms such as WhatsApp offer limited support for modifying or withdrawing access to shared photos. Existing features like message deletion and view-once media provide partial controls but do not support ongoing, revisable consent at the photo level. This project investigates how consent revocation can be designed as a usable interaction within WhatsApp-style photo sharing. Drawing on literature concerning dynamic consent, multi-party privacy, and privacy-focused HCI, four research gaps are identified. In response, Smart Photos is proposed which is a permission model where each shared photo carries a persistent and revisable permission state covering forwarding, downloading, view-once, time limited access, and co-subject governance. A front-end prototype was built with React, Next.js and TypeScript and evaluated through semi-structured interviews with 9 participants. Thematic reflexive analysis used and five themes are extracted which shows that participants valued revisable permissions as a meaningful improvement while recognising that revocation remains inherently partial. The dissertation also discusses recommendations for different stakeholder groups that seeks photo sharing platforms to adopt a similar model.

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Chapter 1

Introduction

The widespread use of digital messaging platforms has transformed how people share personal images, maintain relationships, and negotiate boundaries in everyday communication. Despite developments in security safeguards, users continue to face substantial difficulties in controlling shared content once it reaches another person's device. Current messaging platforms provide protection against external threats but offer limited tools for managing interpersonal risks such as regret, changing relationships, or misuse of personal photos [58]. These vulnerabilities are especially concerning given that consent in digital contexts is frequently dynamic rather than static, and users may seek to modify or withdraw access to shared material over time. While recent studies in privacy, digital safety, and interpersonal communication emphasise the significance of modifiable consent, practical design solutions for major messaging apps remain underdeveloped [2, 26].

1.1 Motivations

This project focuses on WhatsApp as its primary case study because it is the most widely used messaging platform for private photo sharing, with over 3 billion monthly active users [16]. WhatsApp employs end-to-end encryption to protect message content from external access, but this model does not extend to post-delivery control between users. Once an image is sent, the sender loses the ability to influence how it will be used, stored, or redistributed. Most existing work on consent revocation in this space focuses on legal or conceptual frameworks rather than on usable, interface-level solutions within everyday messaging platforms. This project is

motivated by the need to address this gap by examining how WhatsApp can better support ongoing, revocable consent when users share photos. Although WhatsApp is the primary case study, the user needs and interaction principles explored are intended to generalise to other messaging platforms that support private photo sharing and forwarding.

1.2 Scope

This project examines pre-send and post-sharing consent revocation for photos in WhatsApp one-to-one private chats. The primary focus is on how users can modify or withdraw consent before and after sharing photos, and how interface design can support this across different social dynamics. The scope is limited to image-based content and other media types are excluded. The research is confined to user experience and interface design, investigating user-facing interaction patterns rather than underlying system architecture or encryption mechanisms. Large-scale user testing, behavioural studies, and technical feasibility assessments fall outside the project scope.

1.3 Research Questions

This project is guided by a set of research questions that aim to investigate how consent revocation can be effectively supported within WhatsApp's messaging environment. These questions address the limitations of current functionality, the expectations and concerns of users, and the design requirements for implementing revocable consent, shaping the direction of the thesis.

The research questions are as follows (1 main question, 5 supporting questions):

Main research question: How can WhatsApp be designed to support clear, usable, and trustworthy mechanisms for revoking or modifying consent before and after users share photos in private conversations?

1. What limitations exist in WhatsApp's current design for managing photo sharing, and how do these limitations affect users' ability to control shared content?

2. What expectations, concerns, and behaviours do WhatsApp users have regarding post-sharing control of personal photos?

3. Which interface elements and interaction patterns best communicate photo permissions and make revocation options understandable within WhatsApp's existing UI?

4. How can the process of revoking or modifying consent be designed to be intuitive and equal for all users?

5. How do the proposed revocation features affect the' perceived trust, safety, and sense of control of users when sharing photos on WhatsApp?

1.4 Aim and Objectives

The aim of this project is to develop and assess user interface mechanisms that facilitate revocable, ongoing consent within WhatsApp's photo-sharing environment, allowing users to maintain control over shared content in a transparent, intuitive, and reliable way.

To achieve this aim, the project pursues the following objectives:

1. Analyze the limitations of the current WhatsApp design in relation to the control of photo-sharing.

2. Define functional and non-functional requirements for effective consent revocation features.

3. Design a set of interface components and interaction flows that demonstrate how revocable consent can be integrated into the WhatsApp user interface.

4. Develop a prototype that illustrates the full permission and revocation workflow.

5. Evaluate the usability, clarity and trustworthiness of the proposed features through user testing.

6. Reflect on the broader implications for user safety, privacy, and ethical communication design.

1.5 Contributions

The project makes the following contributions:

1. A high-fidelity interactive prototype demonstrating a redesigned consent revocation system for photo sharing in WhatsApp-like messaging environments.

2. A set of realistic interaction flows, UI states, and permission management behaviours modelling how revocable consent could function in practice.

3. A usability evaluation of the proposed design, assessing clarity, perceived trustworthiness, and balance between users.

4. A human-centred design analysis of the practical implications of enabling revocable consent for photo sharing in messaging platforms.

1.6 Key Takeaways

The central finding of this project is that consent revocation in messaging-based photo sharing can be meaningfully supported through interface design, but only partially. The prototype evaluation showed that users valued the ability to revise permissions after sending and recognised it as an improvement over current WhatsApp controls. However, it also showed that revocation is inherently limited by screenshots, external-device capture, and prior viewing, and that users interpret permission changes as socially significant acts rather than neutral technical operations. Effective revocation design must therefore balance control, honesty about limits, and sensitivity to interpersonal dynamics.

Chapter 2

Literature Review

2.1 Brief

This chapter reviews how online messaging platform design affects users' ability to manage consent in photo sharing, with a focus on WhatsApp. It outlines key theories of privacy and consent, examines research on photo sharing and interface design, and evaluates current platform features and identifies the research gaps that justify a user-centred revocation design.

2.2 Digital Messaging and Photo Sharing

Digital messaging has developed from earlier forms of computer-mediated communication into mobile, app-based, and increasingly multimodal communication environments, where text, images, and other media are integrated within the same platforms. Herring's historical overview highlights this shift from predominantly text-based CMC toward convergent and multimodal communication systems, providing a useful background for understanding contemporary messaging platforms [25]. Research on mobile messaging also shows how texting became embedded in everyday social contact, particularly in peer communication shaped by social norms and situational constraints [29]. In parallel, early camera-phone studies show that images were used not only for personal capture but also for communication with absent others and remote tasks, helping establish photo messaging as a routine social practice [32]. This transition is important because it created communication settings in which image sharing carries social meaning and privacy expectations that may later complicate consent revocation.

2.3 Digital Privacy in Modern Communication

2.3.1 Communication Privacy Management (CPM)

Communication Privacy Management (CPM) theory explains privacy as an ongoing process of boundary management rather than a fixed state, making it useful for analysing disclosure in interpersonal and mediated communication [12, 49]. Petronio and Child (2020) describe CPM as a framework in which individuals treat private information as something they own and control, and they manage access to it through privacy rules shaped by context and expectations about appropriate use after disclosure [49]. A key concept is co-ownership which emphasises, once information is shared, recipients may become legitimate co-owners who develop their own assumptions about how that information can be used or circulated [49].

For consent revocation in messaging, CPM highlights a core tension. A sender may understand revocation as reasserting personal control, while a recipient may see it as changing the rules after access has already been granted [49]. This helps explain why post-sharing control is often contentious in messaging environments, where interpersonal expectations shape how users judge trustworthiness, and appropriateness [12, 49]. These tensions are more evident in photo sharing because disclosure is typically associated with continuing relationships. As a result, attempts to withdraw access may be interpreted as relational signals such as mistrust, disappointment, or a reconsideration of boundaries, in addition to privacy management.

CPM's concept of boundary turbulence is relevant here. Turbulence occurs when co-owners hold conflicting privacy rules or when expected information flows break down [49]. In messaging environments such as WhatsApp-style photo sharing, this helps explain why consent may need to be revisited over time and why revocation mechanisms must operate within the tension between social boundary expectations and the technical ease of saving, or redistribution [12, 49].

2.3.2 Contextual Integrity (CI)

Contextual Integrity (CI) reframes privacy as appropriateness of information flows, not simply secrecy or access control [44]. Nissenbaum (2011) argues that information sharing is governed by contextual norms: who is involved (actors), what is shared (attributes), and how it moves (transmission principles) [44]. This is relevant to WhatsApp photo sharing because users often treat a private chat as a bounded context with an implied "right audience," even though recipients may still redistribute the content. Under CI, harm occurs when a photo's flow changes

context (e.g., from a one-to-one chat into a group, another platform, or offline circulation) because the transfer violates the original contextual expectations, even if the sender originally shared willingly [44].

CI is also useful for critiquing the limits of the notice-and-consent paradigm in everyday messaging. Nissenbaum (2011), argues that notice-and-consent does not provide adequate online privacy governance [44]. Likewise, Bourgeois & Vandercruysse (2024) claim that relying too heavily on consent typically results in little or no real privacy control in large-scale data environments [6]. This is important for photo sharing because images frequently involve group privacy: a photo can affect the sender, the recipient, and any depicted co-subjects, each with potentially different expectations about appropriate sharing [6]. In a scenario where photo's are being shared via WhatsApp style means, CI frames revocation as an attempt to restore context-appropriate information flow by governing future access and redistribution.

2.3.3 Media Richness Theory (MRT) and Social Presence

Media Richness Theory (MRT) states that rich media can provide better interpretations of meaning than leaner media, especially when messages contain ambiguity or sensitivity. [13]. Furthermore, Social Presence Theory explains how media vary in making another person feel psychologically present and immediate [55]. Ishii et al. argue that these experiences are shaped not only by channel features but also by evolving social patterns of use [28]. Both frameworks are relevant to WhatsApp-style messaging because it combines rapid exchange with visual content that carries emotional and relational meaning. When users share photos in a medium that feels immediate and personal, they develop stronger expectations about how that content should be handled [13, 55].Therefore withdrawing consent may be experienced not simply as changing access but as a socially delicate act implying mistrust or a shift in relational boundaries. Hence revocation in WhatsApp-style photo sharing needs to be understood in terms of interpersonal meaning as well as technical control.

2.4 The Concept of Consent

2.4.1 The Limits of Formal Consent

Under the GDPR, consent must be freely given, specific, informed, and unambiguous, and withdrawal must be as easy as giving consent; where consent is withdrawn and no other legal

basis remains, the right to erasure may also be engaged [15, 50]. Yet formal compliance does not necessarily make consent meaningful in practice. Baruh and Popescu argue that privacy self-management based on notice and choice places too much responsibility on individuals, even though data practices are often opaque, complex, and difficult to resist in practice [3]. In such settings, consent may be legally valid without being substantively meaningful, creating a validity gap between formal agreement and real user control. They further show that users may respond in two limited ways: through partial withdrawal or avoidance when privacy risks feel unmanageable, or through assimilation, where users accept reduced control and rely on market-provided protections [3].

Waldman extends this critique by showing that privacy is often operationalised narrowly within organisations. In his account, lawyers tend to reduce privacy to notice and choice, while engineers often equate it with security or encryption, leaving limited space for more user-centred understandings of consent and control [62]. For the present topic, this suggests that effective revocation cannot depend only on formal consent or interface disclosure, but must be supported by designs that make control realistic and usable in practice.

2.4.2 Consent Models

Consent in everyday messaging cannot be viewed solely as a single permission granted by an individual, since interactions develop over time through multiple relationships and are often conducted wit a degree of loos of contol. Traditional consent frameworks tend to assume that consent is static, singular, and transaction-like, but this logic becomes unstable in ongoing interpersonal exchange, where the meaning of permission can shift after disclosure. As Burkhardt et al. suggest, meaningful consent depends not only on formal agreement but also on understanding and perceived control, while work on consent and revocation has also shown that decisions made at one moment cannot remain fully informed across the lifetime of data use [1, 10]. This is why dynamic consent has been established as a alternative to static one time authorisation models that ask people to agree to future uses they may neither anticipate nor revisit [54]. In messaging, these weaknesses are particularly visible because permissions can have multiple meanings over time and also may change depending upon how interactions evolve or what occurs subsequent to messages being delivered.

The first weakness is temporal. Static models assume consent is given once and remains valid indefinitely, yet in messaging the meaning of permission shifts as conversations develop. Dynamic

Consent responds to this by treating permission as revisable, emphasising ongoing engagement, granular choice, and adjustable preferences over time [31, 33]. Wilbanks reinforces this by arguing that digital consent should begin an ongoing ethical relationship rather than conclude a one-time administrative step, and later work frames modifiable consent choices as central to user-centred data governance [42, 71]. However, Dynamic Consent improves most clearly the timing of permission, not post-disclosure control. In everyday messaging, being able to revise a preference does not necessarily mean being able to affect content that has already been delivered and moved beyond the sender's immediate control.

The second weakness is that revisability is not the same as withdrawal. Even if a user can update preferences going forward, the harder question is whether permission already granted can be meaningfully taken back. Revocable Consent foregrounds this problem. The literature usefully distinguishes between revocability as a normative requirement and revocation as a practically enforceable outcome. Consent and revocation have been framed as ongoing controls, with informed revocation emphasising that users should understand the forms of withdrawal available to them [1]. Related technical work treats withdrawal as a change in the scope of authorised access where users should be able to specify who may access data, for what purposes, and for how long, while formal models of consent evolution represent revocation by closing a previously open interval of collection or access rather than undoing prior disclosure [37, 51]. This distinction is especially important for photo sharing, where a sender may be able to end future authorised access without being able to retract a photo that has already been viewed, saved, or copied. Revocation in messaging is therefore best understood as limiting future legitimacy and platform-supported access, not as guaranteeing recovery of the original disclosure. This problem is reinforced by the broader notice-and-consent literature, which shows that even ostensibly reversible interfaces can fail to produce meaningful control when the design of choice itself is manipulative or unclear [45].

Two further models address narrower weaknesses. Retroactive consent makes the timing of authorisation explicit, allowing consent and withdrawal to be analysed in terms of whether their effects apply to past or future data practices [51]. Ephemeral consent takes a different approach by limiting persistence prospectively, treating permission as bounded and revisable rather than indefinite [26]. Yet both remain partial. Once content has been copied, remembered, or redistributed by others, neither temporal expiry nor after-the-fact adjustment can fully restore sender control.

Seen comparatively, these models should not be treated as mutually exclusive competitors, but as attempts to repair different weaknesses in conventional consent. Dynamic Consent responds most clearly to temporality; Revocable Consent foregrounds withdrawal and reversibility; retroactive approaches sharpen how timing affects authorisation; and ephemeral approaches seek to limit persistence before problems arise. What unites them is a shared difficulty in accounting for messaging as a social practice in which the meaning and consequences of disclosure continue to evolve after the moment of sending [26, 31, 50, 51]. The central problem is therefore not simply lack of choice, but the challenge of aligning revisability, revocability, relationality, and downstream loss of control within a single model.

For this reason, affirmative and ongoing consent offers the strongest conceptual direction for this dissertation. Im et al. describe affirmative consent as voluntary, informed, revertible, specific, and unburdensome, providing a framework that treats permission as active and revisable rather than presumed or exhausted at a single point in time [26]. These principles fit messaging more closely than one-off models because they better reflect how consent is negotiated in unfolding interaction. At the same time, they should not be overstated as a technical solution to screenshots, retention, or redistribution. Rather than being technological solutions, their conceptual and normative values are in providing an even better understanding of how designers should implement responsible design that allows for permissions that can be revisited, clarified, restricted, and renegotiated; without the assumption that post-sharing control will ever be entirely possible.

This issue becomes even more complex in photo sharing, where consent may extend beyond a single sender-recipient relationship.

Taken together, the literature suggests that the main weakness of conventional consent models is not simply insufficient granularity, but their inability to account for the ongoing social life of shared content. Improved models repair important dimensions of this problem, but they still encounter hard limits once messages and photos enter interpersonal networks where recipients can retain, reinterpret, copy, or redistribute them. For everyday messaging, meaningful consent is therefore best understood not as a completed transaction but as an ongoing and socially situated process whose practical force depends on both relational norms and the technical limits of post-sharing control.

2.5 Consent in Instant Messaging and Photo Sharing

2.5.1 Multi-Party Privacy Conflicts (MPCs)

Multi-party privacy conflicts arise when one photo affects several people with different privacy interests, making individualised consent models inadequate. In the photo-sharing literature, MPCs are generally understood as situations in which the person who decides to share content is not the only person who bears the privacy consequences of that disclosure. One user's act of sharing can expose others depicted in the image, even where those others have had no meaningful opportunity to approve, refuse, or shape that disclosure. Cherubini et al. show that this problem is especially remarkable for co-owned content such as group photos, while Liu et al.'s survey of image privacy similarly suggests that privacy in online image sharing is shaped not only by image content itself but also by relationships, context, and multi-party interaction [11, 35]. Applied to instant messaging, this means that a shared photo may simultaneously involve the sender, the recipient, and any depicted co-subjects, each of whom may hold different expectations about whether the image should be sent, retained, forwarded, or later withdrawn. Consent in photo sharing is therefore not only something that may need to be revisited over time, as argued in the previous subsection, but also something distributed across multiple stakeholders from the outset, which makes sender-only models of permission normatively and practically incomplete [11, 35].

The literature proposes several responses to multi-party privacy conflicts, but these approaches often manage the problem more effectively than they resolve it. Cherubini et al. distinguish between precautionary mechanisms, which prevent or constrain sharing before disclosure, and dissuasion mechanisms, which seek to discourage risky sharing by prompting reflection on possible consequences [11]. They find that participants generally prefer precautionary mechanisms because these provide stronger protection for affected parties and reduce uncertainty for uploaders, although they also introduce friction and depend on identifying co-owners in advance [11]. This limitation is reflected in technical proposals. Xu et al. (2017) propose a system in which people appearing in a photo can be identified and involved in the sharing decision, but this depends on a privacy-preserving facial-recognition process to detect co-owners in the first place [72]. Again in another paper, Xu et al. (2019), similarly develop a trust-based privacy-preserving sharing model that allows selective protection of co-owned content, however the approach still treats the problem largely through pre-sharing optimisation and does not fully incorporate direct negotiation among all affected parties [73]. Zhang et al. (2023) extend this

critique by arguing that privacy-preserving frameworks often define ownership too narrowly and do not fully address the risks introduced by subsequent forwarding and cross-platform propagation [74]. When combined, these methods demonstrate that co-ownership cannot be reduced to sender-side privacy settings. However, they are mostly pre-sharing, system-centered, or prediction-based, and as a result, they suffer when images become technically and socially mobile.[11, 72, 73, 74].

Multi-party privacy conflicts therefore show that consent revocation in messaging is not simply a matter of allowing one sender to change their mind, but a broader problem of shared stakes and weakening control once images begin to circulate [11, 74].

2.5.2 Persistence, Recipient Control, and the Limits of Revocation

Multi-party privacy conflicts show that control over a photo may already be shared before it is sent however the next limit is structural. In instant messaging, revocation does not operate as ongoing access control after delivery, because the system's basic function is to transmit content to a recipient-controlled device rather than to preserve sender authority over it indefinitely. For that reason, revocation is best understood as prospective and partial. Once a photo has been delivered, a platform may still alter later in-app availability, but it cannot guarantee retrieval of the original disclosure. This is consistent with earlier work on revocation as a forward-looking rather than fully retroactive mechanism, and with Schnitzler et al.'s observation that deleting messages from a recipient's history is substantially harder than deleting them locally, even in closed messaging ecosystems [1, 50, 53].

WhatsApp provides a useful case because its privacy architecture and its revocation features point in different directions. On the one hand, WhatsApp emphasises end-to-end encryption and states in its UK Privacy Policy that messages are typically stored on users' devices rather than on its servers once delivery has occurred, on the other hand, this model means that post-delivery control is necessarily limited [66, 70]. Deletion, disappearing messages, and view-once media should therefore be interpreted as bounded controls over later visibility, not as guarantees of erasure. WhatsApp's own help documentation states that recipients may see a message before deletion succeeds, that unsuccessful deletion may not be flagged to the sender, and that iOS recipients may still have media saved in Photos even after the message is deleted from the chat [68]. Likewise, disappearing messages only affect new messages and can be kept in the chat, while unopened view-once media can still be restored from backups. By limiting

export and auto-download, even more recent safeguards like Advanced Chat Privacy are said to help in preventing content from being taken outside of WhatsApp, which emphasizes that these measures reduce rather than completely remove persistence pathways. [64, 65, 67, 69].

The weakness of revocation is not only technical but also interpretive. Schnitzler et al. show that users do not always correctly understand what a deletion action actually does, and that clearer interface explanations improve their ability to assess deletion outcomes; in their study, participants in the WhatsApp condition understood message deletion more accurately than those in the Skype condition, partly because WhatsApp more clearly distinguishes local from broader deletion effects [53]. Furthermore, the same study also shows that deletion notices and other residual traces matter socially. Participants were sometimes surprised not by whether deletion occurred, but by the notification that a message had been deleted, and some explicitly associated such traces with mistrust or ambiguity. In this sense, revocation in WhatsApp photo sharing is also an expectation-management problem. If interface cues suggest stronger control than the system can actually deliver, users may overestimate what deletion has achieved; if revocation is made too visibly traceable, it may also generate interpersonal friction rather than simply restore privacy [53].

2.5.3 The Screenshot Threat

The screenshot threat matters because it creates a form of persistence outside the platform's own retention settings. WhatsApp's privacy-oriented features, including disappearing messages, 'view once' media, and advanced chat privacy, can limit some later visibility or sharing within the app, but they do not fully govern what a recipient can do after viewing content [64, 65, 67, 70]. As Heath et al. note, even disappearing-message systems leave open preservation routes such as forwarding, screenshotting, copying, saving, or photographing content before it disappears [23]. As Ingber argues, screenshots turn part of a conversation into a new file on the recipient's device, making it easier to circulate beyond the original exchange [27]. Screenshotting therefore creates a parallel persistence route that sits partly outside the logic of in-app revocation and ephemerality. Even if the original message becomes less visible within WhatsApp, a copied representation may continue to exist elsewhere. More broadly, image-privacy scholarship identifies the recipient side as a continuing source of privacy risk once content is already in circulation, because recipients may retain or recirculate images beyond the sender's intended audience [35].

Critically, screenshots are not only technical workarounds but socially meaningful acts. Ingber distinguishes screenshot collection from screenshot sharing, showing that both generate distinct privacy harms by moving co-owned information into a new context without notice or consent [27]. Since affirmative consent must be specific [26], permission to share a photo with one recipient does not automatically translate to agreeing it to copy or recirculate. Screenshots therefore make revocation not only technically weak but normatively contested. This means recipients may treat capture as harmless remembering or legitimate reuse, while senders may understand it as a breach of containment and trust. Screenshot risk should accordingly be treated as a central challenge for any interface that presents revocation or ephemerality as meaningful user control [23, 27, 35].

2.6 Human Computer Interaction and Interface Design

HCI research shows that consent and revocation controls often fail in practice when users cannot easily find or confidently use privacy settings in fast-paced sharing contexts [6, 71]. This problem is closely connected to whether privacy options are presented transparently and neutrally, allowing users to act on them meaningfully rather than being guided toward platform-preferred outcomes. [45].

2.6.1 Interface Failures and Manipulative Patterns

Even when platforms formally provide privacy and consent controls, HCI research shows these mechanisms often fail to produce meaningful user control at the interface level. This is an issue that becomes especially consequential in messaging-based photo sharing, where disclosures are often fast, and difficult to reverse once an image leaves the sender's device. Classic notice-and-choice approaches are frequently ineffective because notices are not usable or useful in practice, are often ignored, and are frequently treated as something bolted onto a system rather than integrated into its interaction design [52]. In messaging and social media settings, this becomes visible as concrete usability failures. Expert evaluations using privacy-focused heuristics find that the ability to revoke consent is among the most problematic areas across popular apps, and that privacy information is inconsistently placed in menus in ways that undermine core usability principles such as consistency [2]. For photo sharing, the implication is straightforward: users cannot withdraw consent meaningfully if image-related controls (e.g., deletion, visibility, sharing restrictions) are hard to locate, hard to compare, and ambiguous

about what they actually change—particularly when users are acting quickly in-chat and assume a "delete" action restores privacy.

This problem is intensified when interface designs do not merely confuse users but actively steer them toward platform-preferred outcomes. Empirical work on consent dialogs shows that "consent" interfaces commonly use design tactics that encourage acceptance of defaults (e.g., making "accept" more prominent, hiding configuration behind additional steps...), so the banner functions more like a notification than a genuine control mechanism [5]. Analyses of GDPR-oriented consent pop-ups likewise argue that dark-pattern strategies can be used to circumvent the regulation's intent "by design," and that rights-protecting regulation is weakened if interface design is left unaddressed [56]. From a legal/HCI perspective, this steering is especially problematic because valid consent requires a meaningful choice and the ability to withdraw consent, and designs that make "accept" more salient than "reject" or that hide privacy-relevant options undermine that balance [20]. Experimental evidence also suggests that cookie-consent regimes often do not work as intended: participants frequently select privacy-unfriendly options and report low perceived control, raising concerns that interface-level nudges and defaults can hollow out the ideal of deliberative consent [21]. For WhatsApp-style photo sharing, the key implication is that revocation controls must not only exist but must be reachable and resistant to interface choices that overstate user agency or make withdrawal harder than acceptance.

2.6.2 Interactive Design for Meaningful and Revocable Consent

Constructive HCI work suggests that meaningful consent is best supported when it is designed as an explicit, understandable interaction rather than a frictionless checkbox event. Wilbanks argues that in e-consent, interface design shapes whether people understand what they agree to, so consent should promote comprehension rather than mere completion. [71]. Relatedly, affirmative-consent work, frames consent as requiring conditions such as being informed, specific, and revertible, which translates at the interface level into making actions intentional, consequences legible, and outcomes visible [26]. On mobile devices in particular, alternative consent designs that go beyond a standard tap-through can increase attention to what is being agreed to [34]. For photo sharing, these insights imply that consent-relevant controls should be embedded into the moment of sending in a way users can notice under time pressure, and should provide clear feedback so users can tell what has changed after they act.

The same interactional logic extends from initial permission to later revision and withdrawal.

Schaub et al. emphasise that the effectiveness of privacy notices depends on design dimensions such as timing and delivery context, supporting the use of just-in-time and contextually placed cues rather than relying on static policies [52]. Where revocation is difficult to operationalise after disclosure, HCI work on multiparty privacy conflicts shows that design can intervene earlier through precautionary mechanisms (e.g., requiring collaboration/approval) or dissuasive mechanisms (e.g., friction or warnings that prompt reconsideration) [11], these approaches are especially relevant to photo sharing because they can reduce avoidable disclosures before images become socially and technically mobile. Dynamic-consent literature strengthens this orientation by treating permissions as reviewable and updateable over time via interface-supported communication and preference management [31, 33, 54]. Technical proposals for dynamic consent management further highlight capabilities such as explicit consent choices, access to consent history, and withdrawal support across a data lifecycle [37], hough adoption-focused reviews also note practical barriers such as consent fatigue and implementation complexity [33]. For WhatsApp-style photo sharing, the design takeaway is not to promise perfect post-sharing control, but to build interactions that help users act deliberately at sharing time, understand the practical scope of revocation, and revisit decisions later through clear controls.

2.7 Empirical Usability of Current Platforms

Empirical evaluations suggest that revocation-related privacy controls are frequently available in principle but weak in usable effectiveness. In a STRAP-based expert heuristic evaluation of WhatsApp, Twitter, and Snapchat, Albesher and Alhussain identify the "ability to revoke consent" as the most notable problem area across all three apps, noting that revocation is conditional or unreliable and that privacy settings are presented inconsistently, undermining basic usability principles [2]. Complementary legal-empirical evidence from the web consent ecosystem reinforces this availability–effectiveness gap: Kancherla et al. find that many popular sites make revoking consent harder than accepting it, and often fail to properly record or share withdrawals, allowing tracking to continue even after users attempt to withdraw consent. [30]. Revocation therefore tends to be a high-severity usability weakness, not simply a missing feature.

WhatsApp illustrates this concretely. Platform features such as message deletion, disappearing messages, view-once media, and chat-level protections can reduce some forms of persistence, but they operate as mitigations rather than guaranteed erasure [64, 65, 67, 68, 70]. Foren-

sic analysis by Heath et al. further shows that disappearing-message behaviour varies across platforms and storage locations, that content can be preserved through replying, forwarding, or screenshotting before expiry, and that message timers and residual artefacts can remain in application databases [23]. Overall, existing platforms provide controls that gesture toward revocation, but empirical and platform evidence suggests these controls are weaker in practice than their presence implies.

2.8 Research Gap and Summary

Taken together, the literature reviewed in this chapter shows that consent revocation in messaging is neither a purely technical function nor a fully solved interface problem. Earlier sections establish that revocation is socially negotiated, only partially effective after delivery, vulnerable to forwarding and screenshots, and difficult to translate into interaction models that users can understand and act on in practice. The remaining research gaps therefore concern not whether revocation matters, but how it can be operationalised, scoped, and evaluated in WhatsApp-like photo-sharing contexts.

2.8.1 RG1: Translating dynamic/ongoing consent into usable messaging UI

The chapter has shown that dynamic, ongoing, and revocable consent are conceptually better suited to messaging than one-off authorisation, because they treat permission as revisable over time rather than exhausted at the moment of sending [26, 31, 33, 54]. However, the literature reviewed here provides much stronger conceptual and governance-level accounts than concrete guidance on how such models become low-friction, intelligible interaction flows within routine messaging. The unresolved issue, therefore, is how to express ongoing consent through usable controls and in-context permission management for everyday photo sharing without overwhelming users or overstating what such controls can achieve [37, 71].

2.8.2 RG2: Post-delivery control is partial and easily misunderstood under real redistribution conditions

The chapter also shows that real platforms already offer deletion and ephemerality features, but that these operate only as bounded post-delivery mitigations rather than enforceable recall. In WhatsApp-like systems, revocation becomes structurally weaker once content reaches recipient-

controlled devices, and users do not always interpret deletion cues or feature scope accurately [50, 53, 65, 67, 68]. This problem intensifies under actual redistribution conditions, because forwarding, saving, screenshotting, and other capture practices create persistence routes beyond platform-scoped revocation [23, 27]. The remaining gap is therefore not simply that current controls are imperfect, but that there is still no robust model for post-delivery control that remains user-comprehensible once circulation extends beyond the original in-app exchange.

2.8.3 RG3: Multi-party privacy and fairness are under-supported in revocation interaction design

Throughout this chapter, privacy has been described as relational, context-specific, and often multi-party, so, once a photo is shared, senders, recipients, and depicted co-subjects can all have legitimate but competing stakes in how it is retained or circulated [11, 35, 44, 49]. However, existing messaging platforms largely encode revocation as a sender-centred action, offering little support for co-subject agency, negotiation, or explanations that make revocation clear to all affected parties. Therefore, the unresolved issue is about equally distibuting the control power between stakeholders. The literature currently lacks a clear interaction model for revocation that balance stakeholder transparency while minimizing unnecessary interpersonal conflict as well as social sensitivity when exchanging such information. [11, 27].

2.8.4 RG4: Limited WhatsApp-specific empirical understanding of revocable photo-permission expectations

The reviewed literature gives a strong account of why WhatsApp is an important case: its controls are widely used, bounded in scope, and socially meaningful in ways that affect trust, interpretation, and downstream sharing [27, 53, 64, 68, 70]. It also provides platform documentation, heuristic usability evaluation, and evidence about screenshot or deletion effects [2, 23, 27, 53]. What remains under-evidenced, however, is what WhatsApp users themselves expect from revocable photo permissions in ordinary relational contexts. This includes what forms of control they would understand, trust, accept, or regard as socially fair. The final gap is therefore empirical and context-specific, centring on user expectations rather than on the technical existence of platform features.

2.8.5 Summary

Together, these gaps show that the central problem is not simply whether revocation exists, but whether it can be translated into interaction models that are understandable and grounded in the expectations of WhatsApp users. They therefore motivate the need for a messaging-appropriate mechanism that supports usable, socially intelligible, and realistically scoped revocation of photo access in WhatsApp-like contexts.

Chapter 3

Requirements

3.1 Brief

This chapter defines the requirements for the WhatsApp-like prototype developed in this dissertation. Drawing on the literature review, it identifies the relevant stakeholders, use cases, functional requirements, and non-functional requirements for a prototype centred on photo permissions and revocation.

3.2 Stakeholders

1. **Photo Sender:** The sender shares the photo and configures its permissions. This stakeholder requires clear pre-send and post-send controls, visibility over the current state of a shared photo, and interaction patterns that allow revocation or permission change without unnecessary confusion.

2. **Recipient:** The primary recipient receives the sender's photo in the original chat. This stakeholder requires clear feedback about whether a photo remains visible, or whether any kind of revocation action has been taken.

3. **Secondary Recipient:** The secondary recipient appears only when the primary recipient forwards a photo. This stakeholder is necessary because the prototype examines whether sender-defined permissions continue to shape access after redistribution within the platform model.

4. **Co-Subject:** The co-subject is a person depicted in a photo. This stakeholder is necessary because the prototype examines whether co-subjects can request changes to the photo's permissions.

3.3 Use Case Scenarios

(UC-S0): Sender Configures Permissions Before Sending

1. The sender opens a chat and selects or uploads a photo.
2. The sender prepares the photo for sharing by configuring forwarding, view once, downloading, expiry and co-subject declaration.
3. The sender sends the photo into the chat.
4. The photo appears in the conversation with the chosen permission state applied.

(UC-S1): Sender Revises Permissions After Sending

1. The sender clicks the photo options button in the top corner of the shared photo message.
2. The sender selects 'Revocation permissions' from the in-chat action menu.
3. The system displays the available permission controls in the same in-chat menu.
4. The sender revises one or more permissions (e.g. revoking or restoring access, allowing or blocking forwarding and downloading, or setting, changing, or removing a time limit).
5. The system updates the photo state in the relevant stakeholder's chat.

(UC-R1): Primary Recipient Encounters Active, Revoked, or View-Once Access

1. The primary recipient views the photo's current state in the chat interface.
2. The system presents the photo according to its current permission state, so that the recipient either sees the photo, opens it as view-once content, or is informed that access is no longer available.
3. The recipient may open the photo options menu and acts within the limits of the available permissions, such as replying, forwarding, downloading, or requesting access.

(UC-R2): Primary Recipient Forwards an Allowed Photo

1. The recipient attempts to forward the photo.
2. The system checks whether forwarding is currently permitted by the sender's settings.
    - If forwarding is allowed, the forwarding option remains available.
    - If forwarding is not allowed, the forwarding path is blocked.
3. The photo is forwarded with the sender-defined constraints preserved in the prototype model.

(UC-SR1): Secondary Recipient Receives Constrained Forwarded Content

1. The secondary recipient receives the photo in a separate forwarded chat after the primary recipient forwards it.
2. The system displays the forwarded photo using the original sender's current permission state.
3. The secondary recipient views the forwarded content through the interface.
4. The secondary recipient can interact with the photo only within the inherited constraints.
5. If the sender later revokes access or the photo expires, the forwarded content also becomes unavailable in the secondary recipient's chat.

(UC-C1): Co-Subject Requests Revocation of a Shared Photo

1. The co-subject opens their view of the chat containing a photo in which they are depicted.
2. The system informs that the co-subject appears in the shared photo.
3. The co-subject opens the photo or views its current access state through the interface.
4. The co-subject opens the photo options menu from the button in the top corner of the photo message.
5. The co-subject selects 'Request revocation' and may add an optional explanation.
6. The system sends the revocation request to the sender for review and response.

3.4 Functional Requirements (FR)

ID	Requirement	Justification	Gap
1. Pre-send consent configuration			
FR1	The prototype must allow the sender to select a photo and configure permissions before sending.	Supports contextual, pre-send consent by enabling the sender to define sharing conditions at the point of disclosure, consistent with literature on dynamic consent, in-context privacy choices, and interface support for notice and control. [26, 31, 52, 54]	RG1
FR2	The prototype must explicitly address screenshot and download risk (e.g. indicate that screenshots cannot be fully prevented).	Treats screenshot and download risk as a central limitation of photo-sharing control, with literature emphasising that users should be warned before sharing that some forms of capture cannot be fully prevented. [11, 23, 27, 52, 65, 67]	RG2
2. Post-send revocation and permission management			
FR3	The prototype must allow the sender to modify or revoke access to a previously shared photo via contextual in-chat, photo-level controls.	Enables post-send revision and revocation through contextual, photo-level controls, consistent with literature emphasising that permissions should remain reachable after disclosure. [1, 2, 37, 50, 52, 53]	RG1 RG2
FR4	The prototype must show a compact permission summary under each shared photo so every user sees the current access state.	Makes permission states visible in-chat; literature emphasises that users do not always understand what access changes mean, so interfaces should present clear, legible outcomes. [2, 26, 52, 53, 71]	RG2
FR5	The prototype must maintain a live consent state per shared photo recording current permission status, expiry, active or revoked access, and applicable restrictions.	Maintains an updatable record of each photo's permission state over time, consistent with literature emphasising that explicit consent choices must remain available for later review and change. [31, 37, 51, 71]	RG1

ID	Requirement	Justification	Gap
FR6	The prototype should provide clear feedback after key permission-related actions.	Makes the outcomes of permission changes visible and interpretable, consistent with literature emphasising that clear post-action feedback reduces ambiguity. [2, 52, 53, 71]	RG1 RG2
3. Redistribution and forwarding control			
FR7	The prototype must constrain forwarding according to the sender's current permission state.	Allows forwarding only when explicitly permitted, consistent with literature emphasising that forwarding changes information flow and should remain subject to the sender's transmission constraints.[11, 27, 35, 44, 74]	RG2
FR8	The prototype must propagate sender-defined permissions to forwarded content within the prototype.	Ensures forwarded content inherits the original sender's permissions, consistent with literature emphasising that transmission constraints should persist across forwarding to reduce secondary privacy loss. [11, 23, 35, 44, 74]	RG2 RG3
FR9	The prototype should make the forwarded context and secondary-recipient view clearly interpretable.	Makes forwarded context and secondary-recipient status clear in the interface, consistent with literature emphasising that forwarding changes audience and context. [2, 11, 35, 44, 52, 53, 71, 74]	RG2
4. Recipient Awareness and negotiated consent			
FR10	The prototype must provide recipients with a neutral inline indication when access is revoked or when permissions change.	Communicates access changes in neutral language, consistent with literature emphasising that revocation notices should manage expectations without creating unnecessary blame or ambiguity. [11, 27, 49, 53, 55, 71]	RG3
FR11	The prototype must allow the recipient to request extended access, with the sender deciding whether to approve or reject that request.	Allows recipients to request continued access and senders to renegotiate permissions, consistent with literature emphasising that privacy boundaries may be revisited through interaction. [12, 26, 31, 33, 49, 54]	RG3

ID	Requirement	Justification	Gap
5. Co-subject and multi-party privacy control			
FR12	The prototype must support multi-party photos by allowing the sender to mark that a photo contains other people and allowing depicted co-subjects to request revocation of a shared photo, with an optional explanation.	Supports multi-party privacy by recognising that shared photos may affect depicted co-subjects as well as senders. Literature emphasises that disclosure decisions in photo sharing involve shared privacy stakes, and that co-subjects should have some agency to renegotiate disclosure. [6, 11, 35, 44, 49, 72, 73]	RG3 RG4
FR13	The prototype should allow a depicted co-subject to receive a notification that they appear in a shared photo when the sender has marked the image as multi-party content.	Supports co-subject awareness by making it visible when a privacy-relevant disclosure involves them. It is consistent with literature emphasising that awareness is a necessary precondition for agency. [6, 11, 35, 72]	RG3
FR14	The prototype should allow the sender to review and respond to a co-subject's request through a dedicated decision interface.	Supports multi-party privacy as a negotiated process by allowing the sender to explicitly review and respond to co-subject requests. Literature emphasises collaborative privacy mechanisms and treats consent as revisable and shaped through stakeholder negotiation. [11, 26, 49, 72, 73]	RG3
FR15	The prototype should provide neutral feedback when a co-subject request leads to changed permissions.	Communicates co-subject related permission changes while minimising interpersonal friction. Literature emphasises that revocation notices are socially meaningful and should be designed to support expectation management and face-saving. [27, 49, 53, 71]	RG3

ID	Requirement	Justification	Gap
FR16	The prototype should show that co-subject-related permission changes can affect forwarding and secondary-recipient access within the platform model.	Supports multi-party privacy beyond the original chat by showing that co-subject-related permission changes continue to affect redistribution and secondary-recipient access. This aligns with the literature that forwarding changes audience and context, and that downstream distribution can intensify multi-party privacy risk. [11, 35, 44, 74]	RG3
6. Honest Communication of Limits			
FR17	The prototype should allow the interface to explain the limits of protection, including that offline capture, screenshots, or prior viewing cannot be fully undone.	Supports realistic protection by clearly communicating the limits of platform-level control. Literature emphasises that revocation is partial and prospective, and that screenshots, offline capture, and prior viewing cannot be fully reversed.[23, 27, 50, 53]	RG2 RG3

Table 3.1: Functional Requirements

3.5 Non-Functional Requirements (NFR)

ID	Requirement	Justification	Gap
1. Usability and clarity			
NFR1	The prototype interface must be easy to understand for first-time users.	Helps users understand the permission model and core controls without prior training or repeated use. Literature on usable privacy and consent interfaces emphasises that controls should be legible, and consistent. [2, 6, 26, 45, 52, 53, 71]	RG1 RG2
NFR2	The prototype should support quick routine sharing while adding only proportionate friction for privacy-sensitive actions.	Ensures that routine sharing stays quick while privacy-sensitive actions receive enough attention. Literature on usable consent and privacy interaction design emphasises that consent flows should remain low-friction in ordinary use, while more sensitive actions may have warnings that encourage reflection without causing fatigue or avoidance. [3, 11, 26, 33, 34, 52, 71]	RG1
NFR3	Consent and privacy controls must present choices in a balanced, and non-manipulative way.	Ensures that users can understand and compare options and make freely made choices. Literature on meaningful consent and dark patterns emphasises that privacy choices should not rely on hidden options, unequal emphasis, unclear notices, or obstructed decline paths. [5, 20, 45, 52, 56, 71]	RG1 RG3
NFR4	The interface must provide readable and distinguishable permission states and controls.	Ensures that users can understand and interpret permission-related information confidently. Literature on usable privacy and consent interfaces emphasises that permission states and controls should be visually legible and clearly distinguishable so users can understand status changes, take actions, and interpret consequences without ambiguity. [2, 26, 52, 53, 71]	RG1 RG2
2. Performance			
NFR5	Permission-related actions and status changes should remain responsive enough for live demonstration in a local browser environment.	Ensures that permission changes appear quickly enough to support smooth interaction during use and evaluation of the prototype. Literature on usable consent and privacy interaction design emphasises that controls should remain responsive at the moment of action and not introduce disruptive delay. [2, 26, 33, 34, 52, 71]	RG1

ID	Requirement	Justification	Gap
3. Social impact			
NFR6	The interface must use neutral, non-blaming language to reduce unnecessary social tension when permissions change.	Ensures that permission changes are communicated in a socially careful and non-accusatory way. Literature on privacy boundary management and revocation emphasises that such changes are socially meaningful, so interface wording should ambiguity, and interpersonal friction while keeping changes intelligible. [12, 49, 52, 53, 71]	RG3
4. Consistency			
NFR7	Permission menus, indicators, and states should remain consistent with familiar WhatsApp-like interaction patterns.	Ensures that new permission controls are easier to learn and less confusing by following interaction patterns users already recognise. Literature on usable privacy controls emphasises that consistency supports interpretability, and that settings should avoid inconsistent placement or presentation that makes actions harder to locate or understand. [2, 52, 53, 65, 67, 68, 71]	RG1 RG4
NFR8	Permission management should assume continuity with WhatsApp's existing end-to-end encryption model, without requiring changes to the underlying encryption architecture.	Ensures that permission controls remain compatible with familiar secure messaging expectations. This assumption frames permission management as operating alongside, rather than replacing or weakening, WhatsApp's existing end-to-end encryption, and supports consistency by treating it as an additional layer of user control.	RG1 RG4
5. Transparency			
NFR9	The system must not over-promise. It should clearly indicate that revocation may not affect screenshots, saved copies, or prior viewing.	Ensures that users understand the practical limits of revocation clearly and honestly. Literature on revocation and media control emphasises that revocation is only partial and prospective, so interfaces should avoid implying complete erasure where screenshots, saved copies, exports, or prior viewing remain outside full platform control. [23, 50, 52, 53, 70]	RG2

Table 3.2: Non-Functional Requirements

Chapter 4

Design & Specification

4.1 Brief

This chapter presents the design and specification of the proposed consent revocation system for photo sharing in WhatsApp. It translates the requirements and literature findings into a justified interaction design and system specification. The chapter first introduces the Smart Photos concept and design rationale, before covering the specification in section 4.3.

4.2 Design Rationale and Principles

The design of **Smart Photos** is grounded in the central argument of this dissertation: consent in messaging-based photo sharing should not be treated as a one-off permission granted at the moment of sending, but as an ongoing and revisable process shaped by context, relationships, and the practical limits of post-delivery control. In the proposed model, each shared photo is treated not simply as delivered content, but as a photo associated with a persistent permission state that can specify whether it may be forwarded, saved, viewed once, or accessed for a limited period. These permissions are not exhausted at disclosure, but remain reviewable and revisable through the interface after sending. The design principles set out below are informed by Communication Privacy Management, Contextual Integrity, revised consent models, and privacy-focused HCI research [12, 26, 44, 49, 52, 71]. They provide the conceptual basis for the structure of the model and the prioritisation of particular interaction choices.

4.2.1 Dynamic and Revocable Consent by Default

The first design principle is that consent should be treated as dynamic and revocable by default. This follows directly from the literature reviewed in Chapter 2, which shows that one-off consent models are poorly suited to interpersonal messaging, where disclosures unfold over time and may later feel inappropriate as contexts, relationships, or expectations change [26, 31, 33, 54]. In CPM terms, sharing a photo does not eliminate the need for later boundary management; rather, it creates a situation in which privacy rules may need to be revisited and renegotiated after disclosure [12, 49].

For that reason, the design does not treat photo sending as an irreversible transfer of control. Instead, it is structured around the idea that a photo can remain associated with a modifiable permission state, allowing the sender to determine conditions of access before sending and to revise those conditions afterwards. This principle underpins the decision to support both pre-send configuration and post-send permission management in the prototype, rather than limiting user control to a single moment of choice. That also means a photo keeps its original sharing conditions after it is sent, so later edits or allowed redistribution within the prototype do not change the original permissions. In this way, the prototype operationalises the dissertation's first research gap by translating the literature on dynamic and ongoing consent into a messaging-oriented interaction model that remains reviewable after disclosure.

4.2.2 Expectation Management over Absolute Control

The second principle is that the interface should communicate realistic and bounded forms of control rather than implying that revocation can fully undo disclosure. The literature reviewed in Chapter 2 shows consistently that revocation in messaging environments is prospective and partial: once content has been delivered, viewed, copied, downloaded, or captured through screenshots or external devices, no interface can guarantee complete retrieval or universal deletion [23, 27, 50, 53]. This limitation is especially important in WhatsApp-like systems, where platform-level controls can affect later in-app visibility but cannot fully govern what has already been retained on recipient-controlled devices [65, 67, 68, 70].

This principle therefore rejects any design framing that would overstate the application's power. Within the **Smart Photos** model, revocation is treated primarily as a mechanism for modifying or withdrawing future platform-mediated access and redistribution, rather than as a guarantee of complete deletion after delivery. This makes expectation management a core design responsibility. It justifies the inclusion of screenshot and capture warnings, state explanations, and wording that emphasises limited but meaningful control. In practical terms, the goal is to help users form a more accurate mental model of what permission changes can and cannot achieve.

4.2.3 Minimal Social Friction

A third design principle is that revocation and permission change should be communicated in a way that minimises unnecessary interpersonal tension. The literature reviewed earlier shows that revocation is not only a technical event but also a socially meaningful act. Under CPM, changing access after disclosure may be interpreted as a renegotiation of privacy boundaries, while work on social presence and media richness suggests that changes to communication permissions can carry relational signals such as mistrust, distancing, or suspicion [13, 28, 49, 55]. Empirical work on deletion cues further suggests that visible traces such as deletion notices may themselves become socially consequential, sometimes producing ambiguity or discomfort rather than simply clarifying what happened [53].

For that reason, the design aims to reduce friction without hiding the fact that a permission change has occurred. This means using neutral, system-mediated wording and avoiding interface patterns that imply blame. The aim is not to hide revocation, but to present it in a way that it limits the possibility of unnecessary tension. This principle is particularly important for recipient-facing notices and status changes, where the interface must remain clear about changing access while avoiding designs that frame revocation as punitive or suspicious.

4.2.4 Fairness Between Parties

Although the project prioritises sender-side control, the design rationale does not treat consent management as a purely sender-centred matter. Both CPM and Contextual Integrity suggest that once content has been shared, the expectations of other parties also matter, even where the sender retains a legitimate interest in revising access [44, 49]. A sender may view revocation as a reasonable attempt to reassert privacy boundaries, while a recipient may interpret the same action as a change to previously granted access. This tension does not mean the sender should lose the ability to revise permissions, but it does mean that recipient-facing states and notices must be understandable. Although the immediate interaction in the prototype centres on the sender and both types of recipients, the same concern with fairness also extends to photos that involve depicted co-subjects whose interests may be affected by later access or redistribution.

The design therefore seeks to balance sender agency with recipient understanding. Recipients are not treated as passive endpoints who simply lose access without explanation. Instead, the system provides clear indications of what has changed. It also supports limited request-based interaction, allowing the recipient and the co-subject to ask for renewed or revoked access while leaving the final decision with the sender. In this way, the design reflects the literature's view of consent as a socially negotiated process and responds to the need for revocation mechanisms that remain socially intelligible.

4.2.5 Consistency with WhatsApp's Existing Interaction Patterns

The final principle is that new revocation and permission features should remain consistent with the interaction patterns users already associate with WhatsApp-like messaging. HCI research reviewed in Chapter 2 shows that privacy controls often fail not only because they are absent, but because they are inconsistently presented, or difficult to interpret in fast-paced use [2, 52, 71]. In messaging contexts, where actions are often taken quickly and with limited deliberation, recognisable interaction patterns are especially important for first-use clarity and adoption.

For that reason, the design does not attempt to invent an entirely separate privacy-management environment detached from ordinary messaging behaviour. Instead, it follows familiar WhatsApp-like interaction conventions such as photo-level action menus, recognisable icons, in-chat state cues, and lightweight bottom-sheet style controls. This principle supports learnability and consistency, and helps ensure that the proposed system feels like a plausible extension of an existing messaging platform rather than an unfamiliar parallel application. By embedding consent and revocation controls within established patterns of use, the design aims to support meaningful control without introducing so much novelty or friction that the feature becomes difficult to adopt in everyday communication.

4.3 Functional Specification

4.3.1 Pre-send Permission Configuration

To achieve **FR1** and **FR2**, the prototype places photo-permission controls directly within the sender's sharing flow rather than treating privacy as something that can only be addressed after a photo has already been sent. From the sender's chat interface, the user can open a photo picker, choose an image from media library, device files, or camera input, and then either send it immediately with default settings or open a dedicated composer for permission-based sharing. This keeps permission decisions close to the moment of disclosure while preserving a lightweight path for routine sharing.

When the sender chooses to send a photo with revocable permissions, the prototype opens a dedicated pre-send composer. Within this composer, the sender can define the photo's initial permission state before it enters the chat. It allows the sender to set forwarding, view-once, downloading, co-subject marking, and time-limited permissions. Time-limited access is configured through a duration selector. Also, the sender can add a caption before sending. In this way, the act of sending, function as a creation of a photo message with an attached permission state.

To achieve **FR2**, the pre-send flow also includes an explicit risk cue that communicates the limits of later control before disclosure takes place. The composer shows a short warning that screenshots and saved copies cannot be undone. This does not attempt to prevent screenshotting or local retention

technically but instead, it frames such risks as structural limits of platform-level control and makes them visible before the photo is shared. As a result, the pre-send interaction supports more informed permission-setting by combining configurable restrictions with an explicit reminder that later revocation cannot fully reverse all forms of recipient-side capture.

4.3.2 Post-send Permission Management

To achieve **FR3**, the prototype keeps permission management reachable after a photo has already been delivered through contextual, photo-level controls within the chat. Each sent photo includes a small action button in the top corner of the message bubble which opens an in-chat menu. The sender can access both general photo actions and a dedicated revocation-permissions view. Through these post-send controls, the sender can revoke or restore visibility, allow or block forwarding, allow or block downloading, and add or remove a time limit for an existing photo. In this way, post-send control is treated as an ongoing interaction tied to the individual photo message.

To achieve **FR5**, these changes operate on a live consent state associated with each shared photo. Because these permissions remain attached to the same source photo after sending, the prototype can update the state of an existing message rather than replacing it with a new one. The adjacent permissions panel reinforces this model by showing the currently selected photo together with its current status, allowing the sender to inspect the active state before making further changes.

To achieve **FR4**, the prototype also renders these permission states clearly within the chat interface for all relevant participants. Beneath each photo, the interface displays compact state labels which indicates the current permission state of the photo. When access is revoked, the recipient-facing interface replaces the visible photo with an inline placeholder indicating that the photo is no longer available in the current chat. When a photo is configured as view-once, recipients open it in a full-screen view that becomes unavailable after closing. These states within the chat interface make permission changes legible in the conversation.

To achieve **FR6**, key permission-related actions are accompanied by lightweight but persistent feedback within the same interface. After sending, the sender is shown a short summary of the most recent actions in a separate panel. More generally, the outcome of permission changes is reflected immediately through updated state chips under the photo and any visible change in recipient-facing presentation, such as revoked placeholders or expiry countdowns. Instead of using separate confirmation dialogs, the prototype makes post-send actions understandable by updating the same chat and control interfaces used to manage the photo.

4.3.3 Recipient Experience and Access Request

To achieve **FR10**, recipient-facing state labels use neutral, system-like language so that revocation is communicated as a change in access rather than as blame.

To achieve **FR11**, the prototype also allows the recipient to request renewed access when a previously shared photo is no longer available. The recipient can request access with an optional explanatory message from the action menu of the revoked photo. The request does not automatically restore access but, instead, it is routed to the sender side of the prototype as a pending request associated with the source photo.

The sender can then explicitly accept or reject that request. If the sender accepts, the prototype restores the photo's access state for the corresponding source photo. If the sender rejects, the recipient is shown a short follow-up message indicating that the access request has been rejected. In this way, the prototype treats access renewal not as an automatic technical reversal, but as a negotiated interaction in which the recipient can ask for renewed access while the sender retains control over the final decision.

4.3.4 Forwarding and Redistribution Control

To achieve **FR7**, **FR8** and **FR9**, the prototype treats forwarding as a permission-dependent action rather than as an unrestricted continuation of ordinary message sharing. A photo can only be forwarded when the sender's current permission state allows redistribution and the photo remains active. In the chat interface, this is reflected through the photo-level action menu. When the forwarding is blocked, the forwarding option is disabled and when it is permitted, the photo can be forwarded into the secondary-recipient chat. In this way, sender also has a role in managing redistribution.

To achieve **FR8**, forwarded content does not create a separate copy with an independent permission model. Instead, the forwarded message remains linked to the original photo through the same source-photo identifier and inherits the sender-defined permissions as represented in the interface. This means that forwarded content continues to reflect the original photo's visibility status and other active restrictions. If the sender later changes the permission state of the original photo, those changes are also reflected in the forwarded instance associated with that same source photo. The prototype therefore models forwarding as continued circulation under the original permission logic.

To achieve **FR9**, the prototype presents forwarded content in a separate secondary-recipient chat which makes the evaluation easier and helps distinguishing the original recipient's conversation from the downstream one. Within that forwarded view, the interface also indicates that restrictions from the original chat still apply, reinforcing that redistribution remains governed by the sender-defined permission state.

4.3.5 Co-Subject and Multi-Party Privacy Control

To achieve **FR12** and **FR13**, the prototype extends the permission model beyond the sender/recipient pair by allowing the sender to mark, at send time, that a photo contains a co-subject. This is done through the pre-send composer, where the sender can enable a co-subject tag as part of the photo's initial permission state. Photos marked in this way then become visible within a dedicated co-subject view, where the depicted person is shown a clear indication that they appear in the photo. Rather than treating the image as sender-only content, the prototype therefore represents certain photos as multi-party disclosures whose privacy implications extend beyond the original sender and recipient. To achieve **FR13** more specifically, the co-subject is shown a dedicated card stating that they are depicted in the photo and may request revocation, making their involvement explicit.

To achieve **FR12** and **FR14**, the prototype also provides a co-subject request flow through which the depicted person can ask the sender to revoke the photo. This request is initiated from the co-subject's photo-level action menu and opens a small modal in which the co-subject can send a revocation request together with an optional message. The request is then routed to the sender side of the prototype as a pending item associated with the relevant source photo. Revocation is not applied automatically. The sender is given an explicit accept-or-reject decision interface in the sender-only message area. This keeps co-subject privacy management within a negotiated interaction model rather than reducing it to hidden system behaviour.

To achieve **FR15**, the outcome of a co-subject request is communicated through clear but non-blaming feedback. On the sender side, accepted and rejected requests are shown with short outcome labels indicating whether the photo has been revoked or the co-subject has been notified of rejection. If the sender rejects the request, the co-subject receives a follow-up message stating that the revocation request has been rejected. If the sender accepts, the photo's access state is updated without framing the change in accusatory terms. In this way, the prototype makes the outcome of co-subject intervention visible while still aiming to minimise unnecessary interpersonal friction.

To achieve **FR16**, accepting a co-subject revocation request changes the permission state of the source photo itself. Because forwarded copies remain linked to that source and inherit its state, co-subject-triggered revocations propagate into downstream redistribution contexts rather than stopping at the single original exchange.

To achieve **FR17**, the prototype makes the limits of co-subject protection explicit. Both the co-subject view and the revocation-request flow state that protection is in-app only and that screenshots or prior viewing cannot be fully undone, presenting these controls as meaningful but bounded without implying complete erasure after disclosure.

4.3.6 Permission State Flow

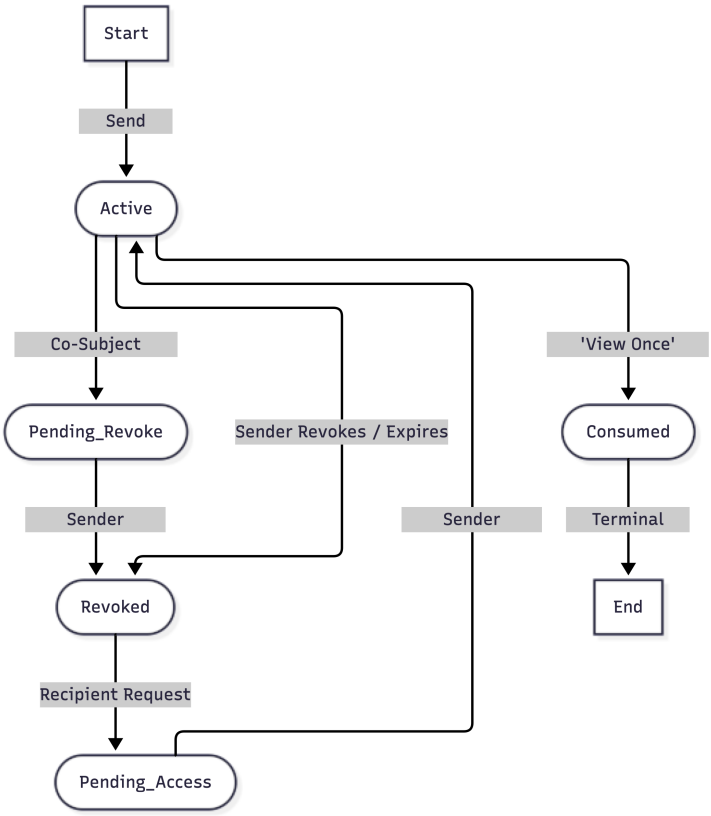


Figure 4.1: Flow chart of Smart Photos

4.4 Non-Functional Specification

4.4.1 Usability and Interaction Clarity

To achieve **NFR1** and **NFR7**, the prototype embeds permission-related controls within familiar, WhatsApp-like interaction points such as the photo picker, pre-send composer, in-chat photo menu, adjacent permissions panel, and message-level state labels. This is intended to make the permission model understandable and easy to learn from first use while keeping its behaviour consistent with current messaging patterns.

To achieve **NFR2**, the interface supports quick completion of routine sharing tasks while introducing additional interaction only where privacy-sensitive reflection is useful. In the current prototype, a selected photo can either be sent directly with default settings or opened in a dedicated composer for permission-based sharing. This preserves a lightweight path for ordinary sharing, while still allowing the sender to configure photo permissions when more control is needed In this way, the prototype aims to introduce less friction.

To achieve **NFR3**, consent and privacy choices are presented as explicit, interpretable options rather than as hidden or one-sided defaults. All pre-send permission controls are clearly labeled buttons in the same viewable panel, which will allow senders to turn on/off each button without having to go through through obscured submenus or being forced toward a single preferred configuration. Similarly, post-send actions are presented as explicit menu choices within the photo-level action flow. This keeps the interface relatively transparent about what options are available and avoids structuring privacy control as a hidden or manipulative interaction.

To achieve **NFR4**, the prototype uses clearly separated action menus, labelled permission buttons, compact state indicators beneath photo messages, and distinct placeholders for revoked content so that users can recognise the current state of a photo without having to infer it indirectly. This applies to all stakeholder views, where visible permission changes need to remain legible after disclosure.

To achieve **NFR5**, permission-related actions remain responsive within the local browser-based prototype so that changes in state appear immediately during interaction and demonstration. In the current implementation, permission updates are applied directly to the active photo state and are reflected at once in the chat view, the permissions panel, forwarded content, and recipient-facing or co-subject-facing states where relevant. This helps preserve a credible and continuous messaging flow, particularly when the sender revises permissions after sending and the resulting changes need to remain immediately visible to other participants.

4.4.2 Socially Careful Communication

To achieve **NFR6**, the prototype communicates all permission changes through neutral, system-mediated language that avoids unnecessary blame or accusation. This is important because revocation, restriction, and access renewal are socially meaningful changes in interpersonal access, not only technical actions. In the current interface, changes are phrased in descriptive terms such as a photo no longer being visible, access having changed, or a request having been accepted or rejected. This principle applies consistently across revoked-photo placeholders, state labels, and request-based feedback, ensuring that permission changes remain visible and understandable while minimising avoidable interpersonal friction.

4.4.3 Consistency, and Honest Communication of Limits

To achieve **NFR8**, the prototype is specified as operating alongside familiar WhatsApp-style secure messaging assumption. The design does not propose any change to the underlying end-to-end encryption model and does not attempt to solve revocation through cryptographic redesign. The specified permission management is framed as an additional interface and state-management layer that governs how photos remain visible or restricted within the platform context. This preserves conceptual conti-

nuity with users' expectations of secure WhatsApp-like communication while keeping the dissertation focused on interaction and consent management.

To achieve **NFR9**, the prototype communicates the limits of revocation explicitly and avoids implying complete erasure after disclosure. In the current interface, this is expressed through pre-send warning text in the composer, explanatory notes beneath shared photos, revoked-state placeholders, and co-subject request flows by stating that screenshots, saved copies, offline retention, and prior viewing cannot be fully undone. The design is intentionally scoped as platform-level visibility control rather than a guarantee of universal deletion. In this way, the prototype remains honest about the partial and prospective nature of revocation and aims to calibrate user expectations about what its controls can and cannot enforce after a photo has been shared.

4.5 Alternative Design Approaches

One alternative design approach considered was a conversation-level privacy mode instead of having photo-level permissions. This approach approach would allow an entire chat to enter a privacy mode in which chared images are blurred by default and restrictions on permission apply across the whole chat.

However, this approach was not selected as the main approach. Firs, conversation-level controls were too broad for the project's central aim of exploring ongoing, revisable consent for photos. Different images within the same chat may require different permissions and durations and a single conversation-level mode could not represent these distinctions effectively. Second concern was about friction in everyday use. A conversation-level privacy mode would require users to think about and menage privacy at the chat level even when only some images are sensitive, which increases the mode confusion and interrupts the ordinary messaging flow. Lee et al. note that dynamic consent systems can face barriers such as consent fatigue and implementation complexity, and a chat-wide privacy mode risked increasing this burden in everyday messaging [33]. For both of these reasons, such design carried a greater risk of being ignored by users over time.

Also at an earlier stage, the project was considering representing the idea of Smart Photos through a high-fidelity Figma prototype. However, this approach was also eliminated. Unlike a Figma-only prototype, a coded version could better represent the permission states of the system adn therefore provide a closer approximation of the intended user experience. In the end, the decision was made to build a coded prototype which will be explained in the next Chapter 5.

Chapter 5

Implementation

5.1 Brief

This chapter details the implementation of the Smart Photos prototype, covering technology choices, the iterative development process, the core architecture and shared state model, a final system walk-through, and testing results.

5.2 Technology Choices and Prototype Scope

Next.js and React

React 19 was chosen for its component-based architecture, which supports modular development, reuse across the interface, and manageable UI state [38]. This suited the prototype particularly well, as the sender, recipient, forwarded recipient, and co-subject perspectives could be implemented as distinct interface regions while still sharing a common permission state. React's built-in state hooks allowed permission changes to propagate immediately across the relevant views, supporting the responsiveness specified by **NFR5**. Because the prototype operates as a single-browser simulation, this behaviour could be implemented without introducing an external state-management library.

Next.js 16 was used as the application framework, providing project scaffolding, routing, TypeScript integration, CSS processing, and an optimised production build pipeline [61]. Although the prototype is essentially a single-page application, Next.js streamlined development and supported rapid iteration across the two implementation rounds described in Section 5.3.

TypeScript

TypeScript was used to define and enforce the core data structures of the prototype at compile time [39]. This was particularly important because the Smart Photos prototype relies on a structured permission model spanning message content, stakeholder roles, and photo-specific control. Several key types were used throughout the implementation:

- `PhotoPermissions` — encodes revocation status, forwarding, view-once, download, time-limit, and co-subject flags;
- `ChatMessage` — a discriminated union distinguishing text messages from photo messages, each carrying different metadata;
- `Participant` — a literal union (`"sender"`, `"recipient"`, `"forwardedRecipient"`, `"coOwner"`) that restricts handler functions to valid stakeholder roles.

Using explicit types helped keep these structures consistent across components and reduced the likelihood of development errors as the prototype evolved. This was especially valuable because the permission model is the central contribution of the prototype and any inconsistency in how permissions, messages, or stakeholder roles were represented would have weakened the credibility of the resulting user evaluation. Although TypeScript does not replace runtime permission logic, it improved maintainability and made the implementation safer to extend during iterative refinement.

Tailwind CSS

Tailwind CSS 4 was chosen as the styling framework because its utility-first approach enabled rapid and consistent interface development without relying on large separate stylesheet files [57]. This suited the prototype well, as the interface needed to visually approximate familiar WhatsApp patterns in order to support the familiarity and consistency specified by **NFR7**. Tailwind's utility classes made it straightforward to replicate these patterns while keeping styling close to the relevant component markup.

The framework was also useful during iterative refinement. Because visual adjustments could be made directly within the JSX, interface details could be revised quickly between implementation rounds without introducing substantial styling overhead. This was particularly valuable in a prototype that underwent multiple rounds of refinement following participant feedback.

Front-End Simulation and Prototype Scope

The prototype operates entirely as a front-end simulation with in-memory state. Conversation data, permission records, and photo metadata are managed through React state in the browser, with no

server, database, or external API. This design was chosen to support the project's aim of evaluating the interaction design and user experience of consent revocation, rather than the engineering feasibility of a production messaging system. It also allowed multiple perspectives to be demonstrated in a single browser window and made permission changes immediately observable during evaluation.

As such, the prototype is an interactive proof-of-concept for the Smart Photos permission model. It demonstrates revocable consent, permission configuration, forwarding control, view-once behaviour, time-limited access, and co-subject involvement, but does not include production infrastructure such as encryption, authentication, persistent storage, real-time delivery, multi-device synchronisation, automated co-subject detection, or server-side access control.

Testing Tooling

Vitest was used as the test runner for the prototype because it integrates naturally with the Next.js and TypeScript development environment and supports fast execution of both logic-level and UI-level tests. React Testing Library was used for rendering components and interacting with them through accessible roles, labels, and visible text rather than implementation details. This was appropriate for the Smart Photos prototype because the key question was whether the interface presented the correct permission state and interaction options to users, not whether particular internal variables changed in a specific way. The tests were executed in a jsdom environment, which provided a simulated browser DOM suitable for unit, component, and integration-style testing of the front-end prototype.

Version Control and Deployment

GitHub was used for version control throughout development, supporting iterative process across the prototype's implementation rounds. The prototype was deployed on Vercel, which provided lightweight hosting for demonstrating the Next.js front-end during testing and evaluation. A link to the deployed version is given at the end of the Section 5.5.

Technology Summary

Technology	Version	Role in Prototype
React	19.2.3	Component-based UI library; manages the multi-view layout and reactive permission state via hooks.
Next.js	16.1.6	Application framework; provides the App Router, development server, and optimised production builds.
TypeScript	5.9.3	Static type system; enforces the permission data model at compile time.
Tailwind CSS	4.2.1	Utility-first CSS framework; enables rapid, consistent styling that approximates WhatsApp's visual patterns.
Vitest	3.2.4	Test runner; executes unit, component, and integration-style tests with coverage via `@vitest/coverage-v8`.
React Testing Library	16.3.2	Renders components in tests and queries the UI by roles, labels, and visible text rather than implementation details.

Table 5.1: Technology stack and component roles

5.3 Development Process

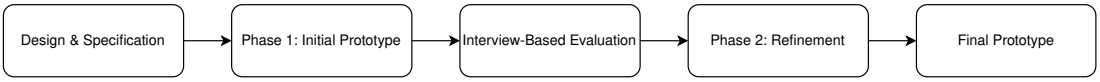


Figure 5.1: Development Process Diagram

Phase 1: Initial Prototype

The first implementation round focused on realising the core interaction flows defined in the functional specification: pre-send permission configuration (FR1–FR2), post-send permission management including revocation and restoration (FR3, FR5, FR6), recipient-side state presentation (FR4, FR10), constrained forwarding to a secondary recipient (FR7–FR9), access-request and response flows (FR11), and co-subject notification and revocation-request mechanisms (FR12–FR17). Each flow was built as a

self-contained interaction path within a single-page React application, with all four stakeholder views rendered side by side so that the consequences of any permission change could be observed across participants in real time.

At this stage the entire prototype resided in a single `page.tsx` file. This monolithic structure was a pragmatic choice for rapid prototyping. This avoided premature abstraction and kept all state and handlers visible in one place while the interaction model was still being tested and revised. That build was then used for the screenshot-driven interviews reported in Chapter 6.

Evaluation and Rationale for a Second Round

A second implementation round was justified on methodological rather than evaluative grounds. As Gould and Lewis argue, usability-oriented design is inherently iterative, requiring a system to be modified and tested repeatedly as it develops from simulation to prototype and beyond [19]. The design in Chapter 4 was derived from literature and requirements and once a runnable prototype existed, it became possible to observe how people interpreted unfamiliar permission states, warnings, and labels under realistic conditions.

For this project, the interviews therefore functioned as a checkpoint between "specification-faithful" code and a version suitable for continued demonstration and assessment. This reflects the role of usability testing in helping designers uncover problems, discover opportunities for improvement, and learn how users actually engage with an interface [40]. A second round was therefore scoped deliberately to layout, styling, copy, and component structure, leaving the core types and permission logic aligned with Chapter 4 unless a change was strictly required for consistency. Chapter 6 documents what was learned from participants and how it was analysed.

Phase 2: Post-Evaluation Refinements

Phase 2 implemented the interface adjustments motivated by that feedback and, in parallel, restructured the codebase from a single file into a multi-component architecture so that further changes and testing could proceed without entangling unrelated parts of the UI.

Architectural Restructuring

During the second implementation round, the prototype was refactored from a single monolithic file into a modular component structure. The original `page.tsx` had grown to contain all rendering logic, state management, and UI concerns in one place, making iterative changes increasingly error-prone. Furthermore, this restructuring improved maintainability and made later refinements more manageable. Splitting the codebase allowed interface changes to be applied to individual components without

risking unintended side-effects elsewhere, while keeping the underlying permission model separated from presentation concerns. The resulting component hierarchy is described in detail in Figure 5.2.

Interface Refinements

Phase 2 focused on improving clarity and prototype fidelity without changing the underlying permission model. Permission information beneath photo messages was condensed into a compact summary with on-demand detail expansion, reducing visual clutter in the chat transcript. The pre-send limitation disclaimer was made more prominent, and the revoked-state placeholder was simplified to reduce repeated explanation. In addition, the prototype was extended to support multi-photo messages with a shared permission set, together with collage and gallery views, bringing the interaction closer to familiar photo-sharing behaviour in mainstream messaging clients. Modal overlays were also scoped to individual handset frames so that each simulated participant view remained visually distinct. Further minor copy and layout adjustments were applied in the same spirit; they are not enumerated here. All these refinements are reflected in the final prototype described in the following sections.

5.4 Components and State Management

Component Hierarchy

The final prototype comprises a thin entry page (`page.tsx`), a central orchestrator component (`PhotoRevocationDemo`), and sixteen supporting components organised into five functional groups. Figure 5.2 shows the hierarchy schematically and a detailed diagram is provided in Appendix X.

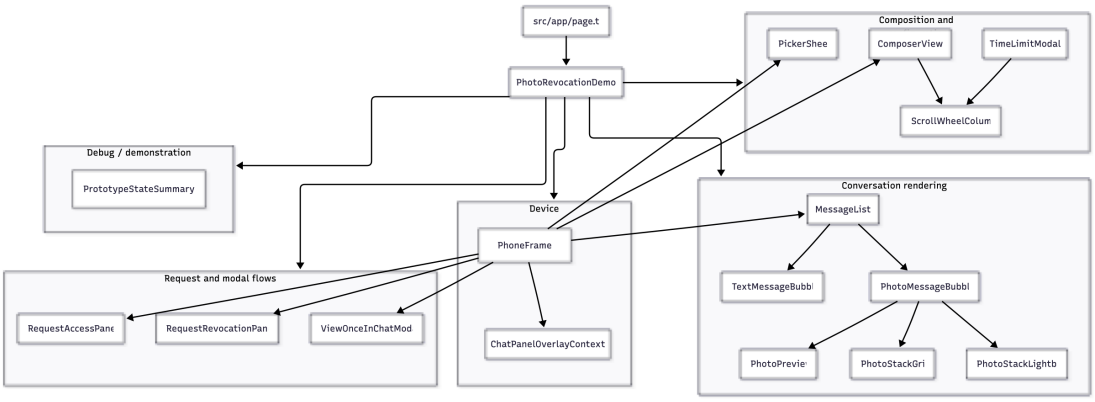


Figure 5.2: Component hierarchy of the prototype

`PhotoRevocationDemo` owns all shared application state and passes data and handler functions down to its children as props. Each simulated handset is wrapped in a `PhoneFrame`, which renders the device chrome and provides a `ChatPanelOverlayContext` so that overlay components such as

ViewOnceInChatModal and PhotoStackLightbox can portal into the correct handset frame rather than covering the entire page. Beneath this shell, the components divide into the following groups:

- **Device chrome.** PhoneFrame renders the handset shell and header for each of the four stakeholder views. It wraps its children in a ChatPanelOverlayContext provider, which exposes a portal target and scroll reference used by overlay components.
- **Conversation rendering.** MessageList iterates over a message array and dispatches each entry to TextMessageBubble or PhotoMessageBubble. Photo messages delegate further to PhotoPreview for single images, PhotoStackGrid for multi-photo collages, and PhotoStackLightbox for full-screen gallery viewing.
- **Composition and configuration.** ComposerView manages the pre-send interface, including caption input and permission toggles. PickerSheet handles photo selection in the sender's footer. Both ComposerView (for the draft time limit) and TimeLimitModal (for post-send adjustment) embed ScrollWheelColumn as a reusable duration picker.
- **Request and modal flows.** RequestAccessPanel and RequestRevocationPanel handle the access-request and co-subject revocation-request interfaces respectively; both are rendered inline in the relevant handset footer. ViewOnceInChatModal manages the view-once confirmation overlay, portalling into the per-handset overlay layer via ChatPanelOverlayContext.
- **Debug and demonstration.** PrototypeStateSummary surfaces the live permission state of the currently selected photo in a panel alongside the handset views, allowing evaluators to verify that permission changes propagate correctly across participants.

Each group corresponds to a distinct area of the interaction model defined in Chapter 4, so that changes to one concern—such as the composition interface—do not entangle another, such as request handling or message rendering.

Shared Permission State

All application state that affects what participants see resides in PhotoRevocationDemo as React useState hooks. The most important of these are 5 message arrays (mainMessages, forwardedMessages, senderOnlyMessages, recipientOnlyMessages, coSubjectOnlyMessages) which together hold every message visible across the four simulated handsets.

The canonical permission record for a photo lives on its entry in mainMessages. Figure 5.3 shows the PhotoPermissions type as implemented in the source (types.ts).

```typescript
export type PhotoPermissions = {
  status: PermissionStatus;
  canForward: boolean;
  viewOnce: boolean;
  expiresAt: number | null;
  canDownload: boolean;
  hasCoSubject: boolean;
};
```

Figure 5.3: The `PhotoPermissions` type with fields matching the sender's permission controls.

Every photo message carries a `sourcePhotoId` that links it back to the canonical entry in `mainMessages`. This linking is encoded in the `ChatMessage` type, which uses a TypeScript discriminated union to distinguish text messages from photo messages (Figure 5.4).

```typescript
export type ChatMessage =
  | {
      id: number;
      sender: Participant;
      type: "text";
      text: string;
      time: string;
      replyToMessageId?: number;
      replyToAssetLabel?: string;
      replyToAsset?: PhotoAsset;
      revocationRequestSourcePhotoId?: number;
      revocationRequestAssetLabel?: string;
      revocationRequestStatus?: "pending" | "accepted" | "rejected";
      accessRequestSourcePhotoId?: number;
      accessRequestAssetLabel?: string;
      accessRequestStatus?: "pending" | "accepted" | "rejected";
    }
  | {
      id: number;
      sender: Participant;
      type: "photo";
      assets: PhotoAsset[];
      caption: string;
      time: string;
      permissions: PhotoPermissions;
      sourcePhotoId: number;
    };
```

Figure 5.4: `sourcePhotoId` links copies to the canonical record.

When the sender revokes, restores, or otherwise modifies permissions, only the canonical entry in `mainMessages` is updated. Forwarded copies in `forwardedMessages` are never rewritten directly. Instead, every rendering component resolves the live permission state at render time by calling `getEffectivePermissions`, shown in Figure 5.5. This function looks up the canonical photo in `mainMessages` by matching `sourcePhotoId`, reads the stored permission fields, normalises `canDownload` and `hasCoSubject` with defaults when they are undefined on the stored record, and applies time-limit expiry logic through `computeEffectiveStatus` before returning the result.

```ts
export function getEffectivePermissions(
  mainMessages: ChatMessage[],
  message: PhotoMessage,
): PhotoPermissions & { canDownload: boolean; hasCoSubject: boolean } {
  const raw =
    mainMessages.find(
      (entry): entry is PhotoMessage =>
        entry.type === "photo" && entry.sourcePhotoId === message.sourcePhotoId,
    )?.permissions ?? message.permissions;
  const expiresAt = raw.expiresAt ?? null;
  const now = Date.now();
  const canDownload = raw.canDownload ?? true;
  const hasCoSubject = raw.hasCoSubject ?? false;
  return {
    ...raw,
    expiresAt,
    canDownload,
    hasCoSubject,
    status: computeEffectiveStatus(raw.status, expiresAt, now),
  };
}
```

Figure 5.5: Resolving live permissions in `helpers.ts`

This derivation pattern means that a single state update to the canonical photo in `mainMessages` is sufficient for every participant view (sender, recipient, forwarded recipient, and co-subject) to reflect the change on the next render cycle. The prototype does not need to synchronise permission records across message arrays or maintain separate update logic for forwarded copies, because every component derives its view of permissions from the same canonical source.

Local Interaction State

Alongside the shared permission state described above, `PhotoRevocationDemo` maintains a second category of state: transient values that control temporary interface behaviour rather than the canonical permission model. These include boolean flags that govern which overlay is currently visible (`pickerOpen`, `composerOpen`, `timeLimitModalOpen`), a structured value tracking which photo menu is active and in which mode (`openPhotoMenu`), and draft text fields that hold in-progress user input before it is committed (`draftCaption`, `accessRequestDraft`, `senderInput`, `recipientInput`). None of these values affect how permissions are stored or derived and they exist only to support the interactive surface of each handset view.

One notable piece of local state is the countdown tick. The prototype supports time-limited photo access, and the remaining time must update visually every second. Rather than introducing a timer event system, the orchestrator increments a counter once per second via a `useEffect` hook, as shown in Figure 5.6.

```
const [, setCountdownTick] = useState(0);

useEffect(() => {
  const id = setInterval(() => setCountdownTick((t) => t + 1), 1000);
  return () => clearInterval(id);
}, []);
```

Figure 5.6: Countdown tick effect in `PhotoRevocationDemo.tsx`. The tick value itself is never read; its sole purpose is to force React re-renders.

The code intentionally discards the state variable during destructuring, as the underlying tick value remains unused. Its only purpose is to trigger a React re-render every second so that `getEffectivePermissions` (Figure 5.5) and `formatExpiryCountdown` re-evaluate with a fresh `Date.now()`. This allows time-limited photos to expire visually in real time and countdown labels to update continuously, without requiring explicit expiry event handling or additional state-management infrastructure.

Separating these transient interaction values from the canonical permission state kept the implementation manageable. The permission model remained single-source-of-truth structure as described in the previous subsection, while the local state layer provided the flexibility needed to support rich interactions such as modal sequencing, draft composition, and live countdown display.

State Transitions: Revocation and Forwarding

The two most consequential state transitions in the prototype are post-send revocation and constrained forwarding. Both operate on the canonical–copy architecture described in Section 5.4, but they update state in different directions. Revocation modifies the canonical record so that all views change, while forwarding reads the canonical record to decide whether a new copy may be created.

Post-Send Revocation

When the sender revokes access, the handler `updateSelectedPhotoPermissions` is called with a partial update (for example, { `status: "revoked"` }). This function maps over `mainMessages`, locates every entry whose `sourcePhotoId` matches the currently selected photo, and merges the update into its `permissions` object, as shown in Figure 5.7.

```tsx
const updateSelectedPhotoPermissions = (updates: Partial<PhotoPermissions>) => {
  if (selectedPhotoId === null) {
    return;
  }

  setMainMessages((currentMessages) =>
    currentMessages.map((message) => {
      if (message.type !== "photo" || message.sourcePhotoId !== selectedPhotoId) {
        return message;
      }

      return {
        ...message,
        permissions: {
          ...message.permissions,
          ...updates,
        },
      };
    }),
  );
};
```

Figure 5.7: `updateSelectedPhotoPermissions` in `PhotoRevocationDemo.tsx`. The immutable map-and-merge pattern updates the canonical record without mutating existing state.

Because `forwardedMessages` is never rewritten, the revocation takes effect across all participant views through the derivation layer already shown in Figure 5.5. On the next render cycle, every `PhotoMessageBubble` calls `getEffectivePermissions`, which looks up the now-revoked canonical record and returns `"revoked"` as the effective status. The same handler is reused for restoring access, toggling the download flag, or enabling the co-subject flag where each is simply a different partial update passed to the same function.

Constrained Forwarding

Forwarding adds a message to `forwardedMessages` only after checking the canonical permission state, as shown in Figure 5.8.

```tsx
const forwardPhotoMessage = (message: PhotoMessage) => {
  const effectivePermissions = getEffectivePermissions(mainMessages, message);

  if (effectivePermissions.status !== "active" || !effectivePermissions.canForward) {
    return;
  }

  const forwardedMessage: PhotoMessage = {
    id: getNextMessageId(
      mainMessages,
      forwardedMessages,
      senderOnlyMessages,
      coSubjectOnlyMessages,
      recipientOnlyMessages,
    ),
    sender: "recipient",
    type: "photo",
    assets: [...message.assets],
    caption: `Forwarded: ${message.caption}`,
    time: getTimeLabel(),
    permissions: effectivePermissions,
    sourcePhotoId: message.sourcePhotoId,
  };

  setForwardedMessages((currentMessages) => [...currentMessages, forwardedMessage]);
};
```

Figure 5.8: Forwarding under effective permissions

Two details are significant. First, the guard clause calls `getEffectivePermissions` to evaluate the canonical record at the moment of forwarding; if the photo has been revoked or its `canForward` flag is `false`, the function returns early and no copy is created. Second, the forwarded message carries the same `sourcePhotoId` as the original. This means that if the sender later revokes access, the derivation layer will resolve the forwarded copy against the updated canonical record, and the forwarded recipient's view will reflect the revocation without any additional update to `forwardedMessages`.

Time-Limit Expiry

Time-limited access follows the same derivation principle without requiring a dedicated state transition. When the sender configures a time limit before sending, the photo message is created with an `expiresAt` timestamp. No handler fires when the deadline passes; instead, `computeEffectiveStatus` (called within `getEffectivePermissions`) compares `expiresAt` against `Date.now()` on every render and returns `"revoked"` once the timestamp has elapsed, as shown in Figure 5.9.

```
export function computeEffectiveStatus(
  storedStatus: PermissionStatus,
  expiresAt: number | null,
  nowMs: number,
): PermissionStatus {
  if (storedStatus === "revoked") return "revoked";
  if (expiresAt != null && nowMs >= expiresAt) return "revoked";
  return "active";
}
```

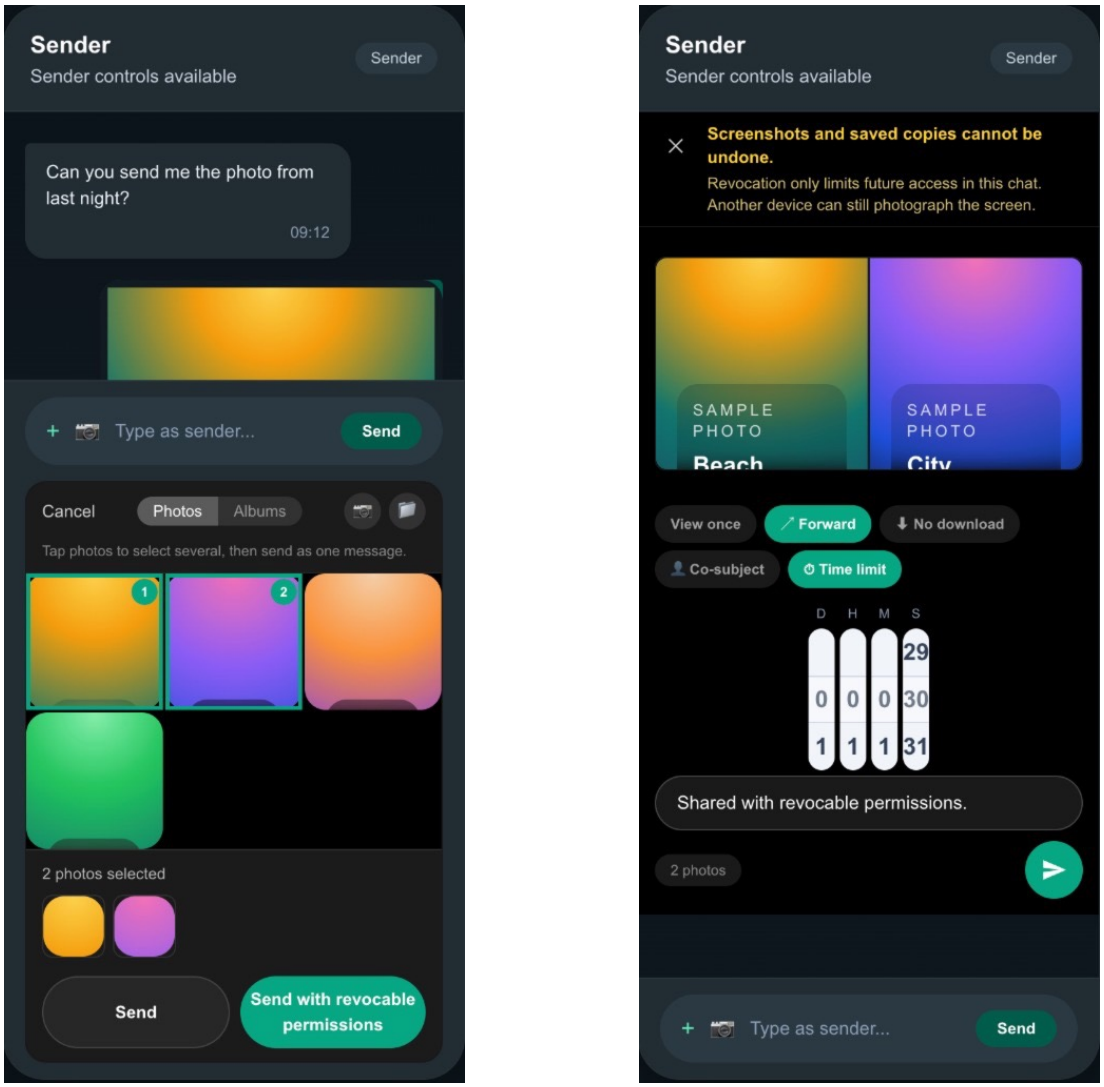
Figure 5.9: `computeEffectiveStatus` in `photoModel.ts`. Expiry is treated as implicit revocation—no event or state write is needed.

Combined with the countdown tick described in the previous subsection, this ensures that a time-limited photo transitions smoothly from an active countdown display to a revoked placeholder at the moment of expiry, using The system translates the concept of ongoing and dynamic consent into a tangible feature by using one core function to handle all access rules. Consequently, any single change made at the source (whether by the user or by time expiring) instantly cascades to update all shared copies of the photo.

5.5 Final Prototype Overview

This section shows how the mechanisms described in Section 5.4 and how the specifications in 4.3 and 4.4 are appear in the final prototype.

Pre-Send Permission Configuration



(a) Photo picker with two images selected.

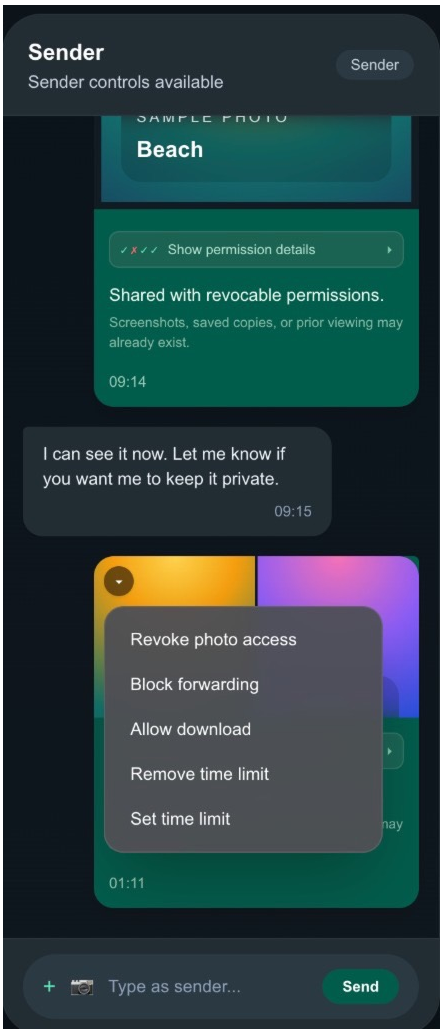
(b) Pre-send composer

Figure 5.10: Pre-send permission configuration flow.

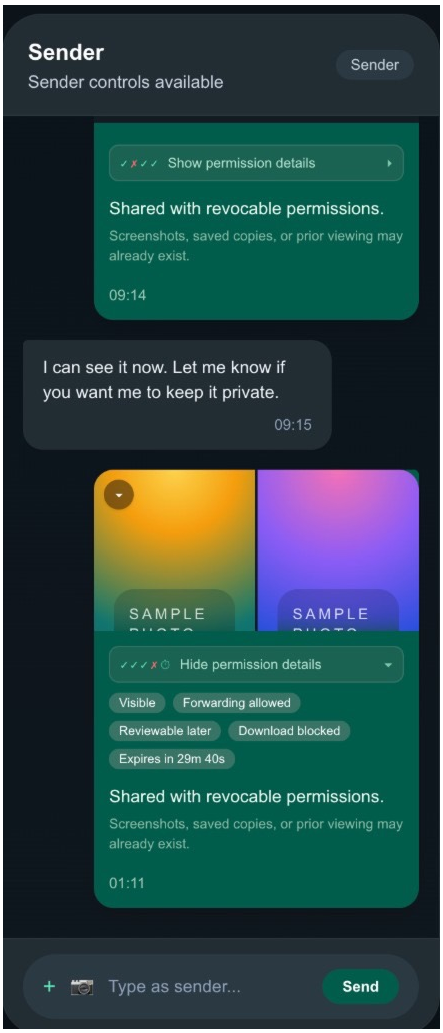
Figure 5.10 shows the realisation of the sharing flow specified in Section 4.3.1. The photo picker (Figure 5.10a) implements the dual-path design as two distinct buttons at the bottom of the sheet. The two options are: *Send* for default sharing and *Send with revocable permissions* for permission configured sharing. Badges numbered on each thumbnail and a summary strip below the grid were added to create an interaction pattern consistent with participant feedback and other familiar gallery selection patterns (NFR7). The pre-send composer (Figure 5.10b) displays the five permission toggles as horizontally arranged chips to keep the composer compact enough to display alongside the staged photos and caption field without scrolling. The FR2 screenshot warning is implemented as a persistent, non-dismissible banner at the top of the composer screenso that it remains visible regardless of the scroll

position. After confirmation has made, the photo is sent to the conversation through the canonical state mechanism described in 5.4 and permission summary of chips will be shown below the message bubble.

Post-Send Permission Management



(a) Revocation submenu on a sent photo, showing the five post-send controls.



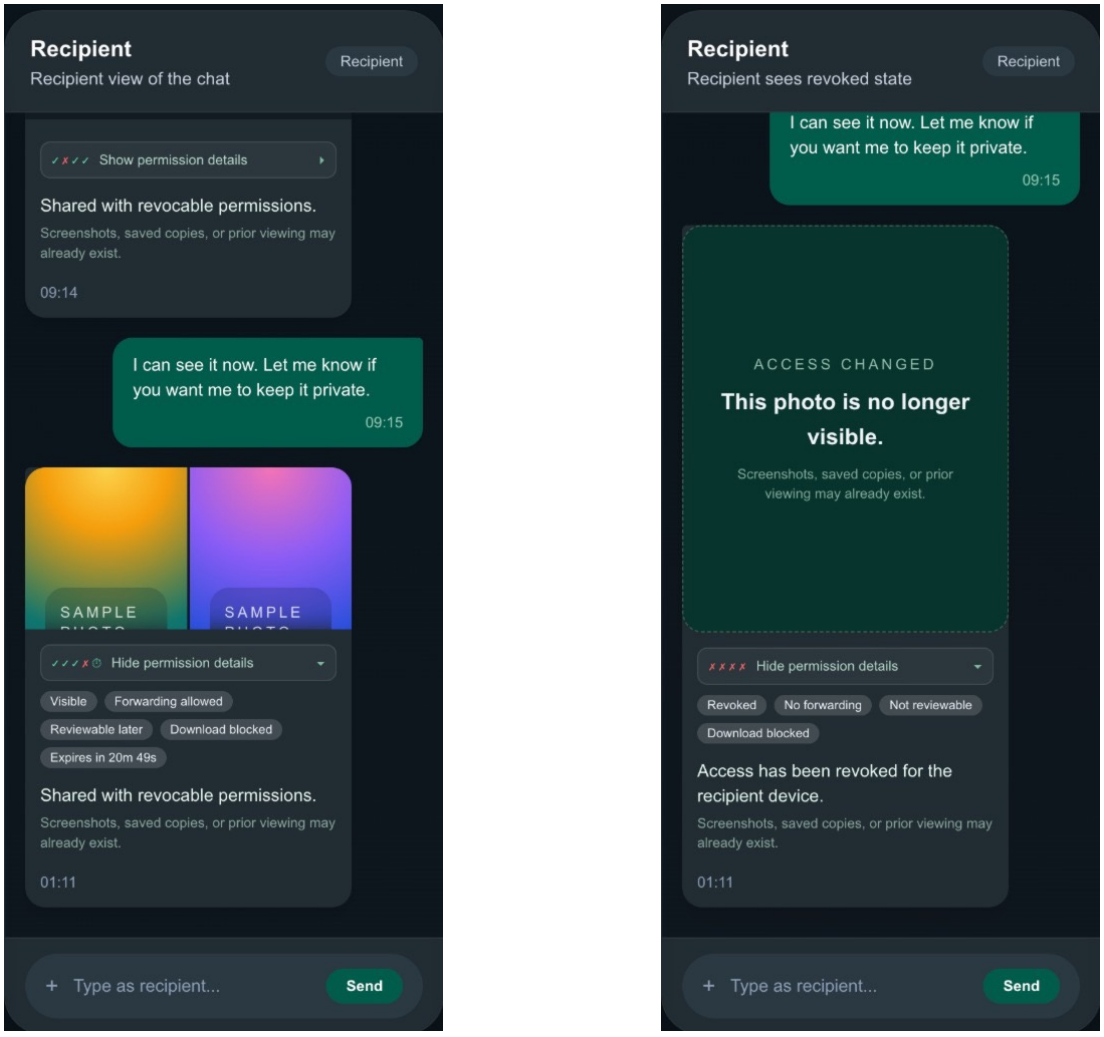
(b) Expanded permission summary under the message bubble

Figure 5.11: Post-send permission management

Figure 5.11 shows how the post-send controls specified in Section 4.3.2 appears. The revocation submenu (Figure 5.11a) is accessed through a drop-down overlay on the message bubble and presents each control as a transition label rather than a current-value label. For example the sender sees *Revoke photo access* rather than *Status: active*, so that available action is alway s immediately legible. This wording convention was an implementation-level decision in order to reduce the cognitive step between reading and acting and not detailed in the specifications. The permission summary beneath each photo

(Figure 5.11) implements the state visibility required by FR4 through a collapsible row. In its compact form, colour-coded tick and cross indicators give an at-a-glance status reading and expanding the row reveals labelled chips such as *Visible*, *Forwarding allowed*, *Reviewable later*, *Download blocked* and live expiry countdown (*Expires in 29m 40s*) that updates every second through the countdown tick mechanism described in section 5.4. The collapsable design was adopted during Phase 2 refinement in response to the interface density concerns raised by participant (discussed in Section 6.3.5), balancing the need for visible permission state against the risk of overwhelming the chat transcript. Because all stakeholder views derive their display from the same canonical record (Section 5.4), a single permission change by the sender is reflected across every stakeholder's bubbles on the next render cycle without additional synchronisation.

Recipient Experience and Revoked States

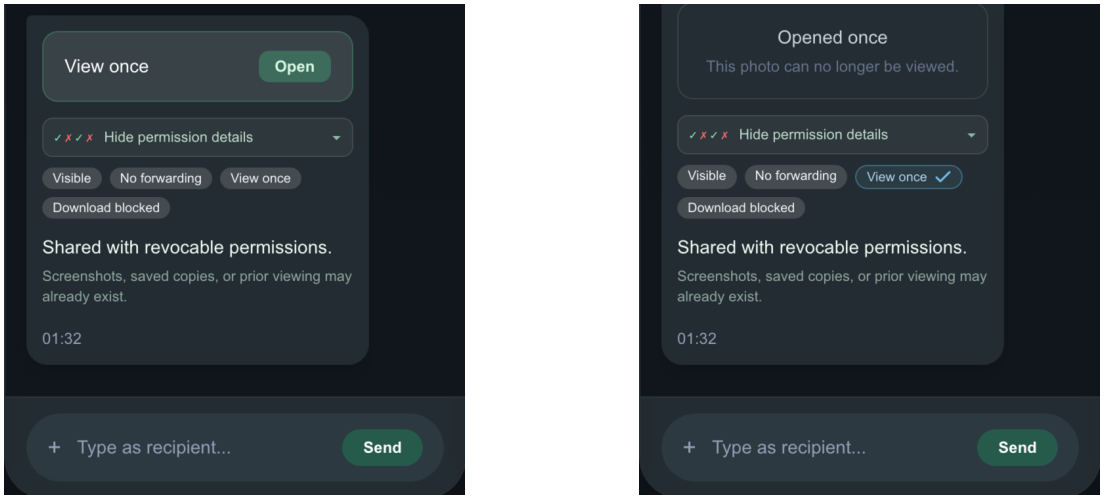


(a) Active photo with expanded permission chips and live expiry countdown.

(b) Revoked state: photo replaced by a neutral placeholder.

Figure 5.12: Recipient experience: active photo (a) and after sender revocation (b).

Figure 5.12 shows the recipient-side realisation of the states specified in Section 4.3.3. The recipient's view reuses the same chip-based summary row as the sender's interface but strips out all management controls, so the action menu is limited to *reply*, *Forward* (if permitted), *Download* (if permitted) and *Request acces* (when revoked). This role-based filtering is handled at render time by checking the user type against each menu option, rather than maintaining separate menu definition per stakeholder. When access is revoked (Figure 5.12b), the photo is replaced by a neutral placeholder reading "Access changed" accompanied by the note that screenshots, saved copies or prior viewing may already exist. The same placeholder is used regardless of whether revocation was manual or triggered by time-limit expiry, since both resolve through the identical derivation path in Section 5.4. This way it ensures the recipient experience remains consistent across all revocation causes and simplifies the language used which minimises the social friction. Even View-once photos has already been implemented in current WhatsApp it is revised. The revision follow a distinct visual appearance shown in Figure 5.13a where a bubble initially displays a "View-once" prompt with an *Open* button and after the recipient opens and closes the full-screen viewer managed by `ViewOnceInChatModal`, the bubble transitions to an "Opened once" notice and the chip gains a tick mark to indicate consumption (Figure 5.13b). The state change is triggered by a "consumed" indicator that activates when the view is closed. Because this represents a final condition, the sender cannot undo it which distinguishes it from regular revocation processes.



(a) View-once photo before opening. (b) After opening, content consumed.

Figure 5.13: View-once lifecycle on the recipient handset: before (a) and after (b) consumption.

Forwarding and Access Requests

Figure 5.14 demonstrates the three-way propagation behaviour specified in Section 4.3.4. Because forwarded copies share the same `sourcePhotoId` as the original message rather than maintaining an independent permisison record, a single state update to the canonical entry in mainMessages is sufficient

to revoke access across all displays. The forwarding guard clause shown earlier in Figure 5.8 enforces the specification's contraint that forwarding is only available when the photo is active and `canForward` is true. So the *Forward* option is disabled dynamically at render time in recipient's action menu. This preserves layout consistency while still communicationg the restriction. At this point the recipient's menu includes *Request access* as the only actionable option beyond *Reply*.

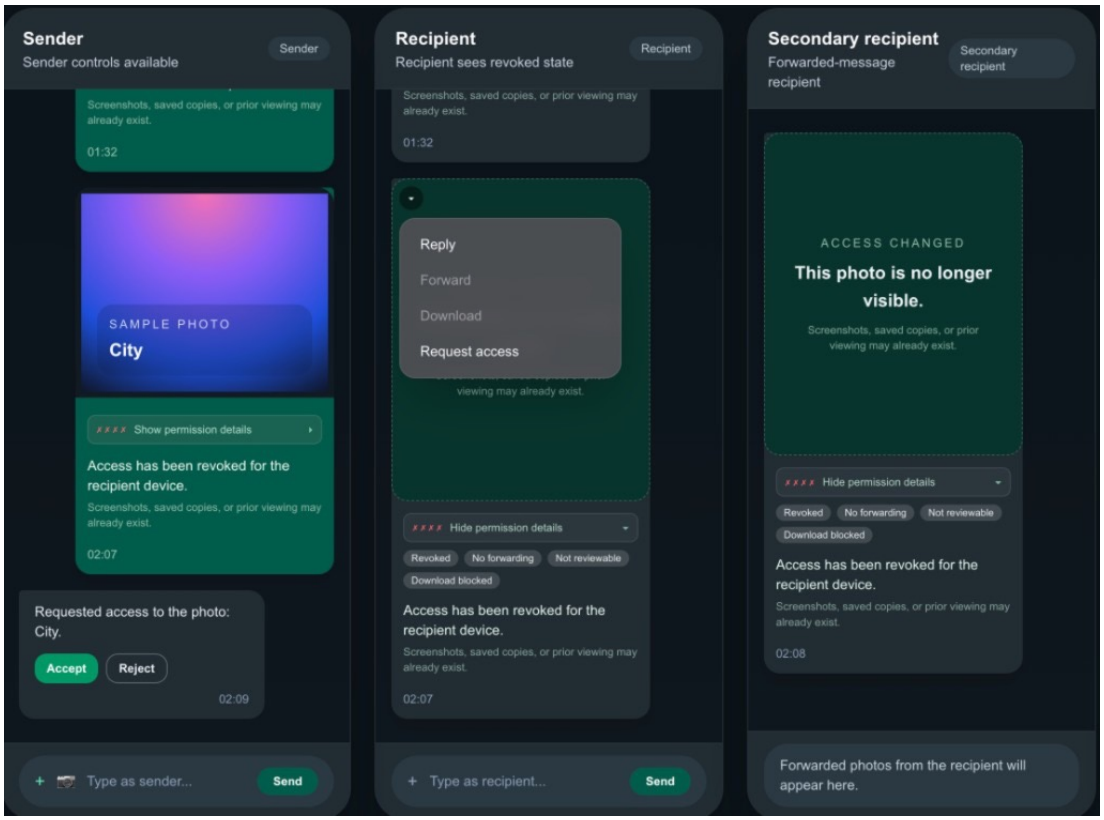


Figure 5.14: Three-view state after sender revocation of a forwarded photo. Left: sender with access-request notification (*Accept*/*Reject*). Centre: recipient with revoked placeholder and *Request access* in the action menu. Right: secondary recipient with propagated revocation.

When therecipient selects *Request access*, the request appears in the sender's thread as a text message carrying the photo label together with *Accept* and *Reject* buttons (Figure 5.14) which implements the negotiated access-renewal flow specified in section 4.3.3. If the sender accepts, the canonical recors is restored to active and all views return to showing the photo with active permission chips as shown in Figure 5.15. The side-by-side layout used in both figures was a deliberate evaluation choice that allowed participants to observe how a single permission change propagates across downstream recipients in real time, making the single-source-of-truth architecture described in Section 5.4 directly verifiable through the interface itself.

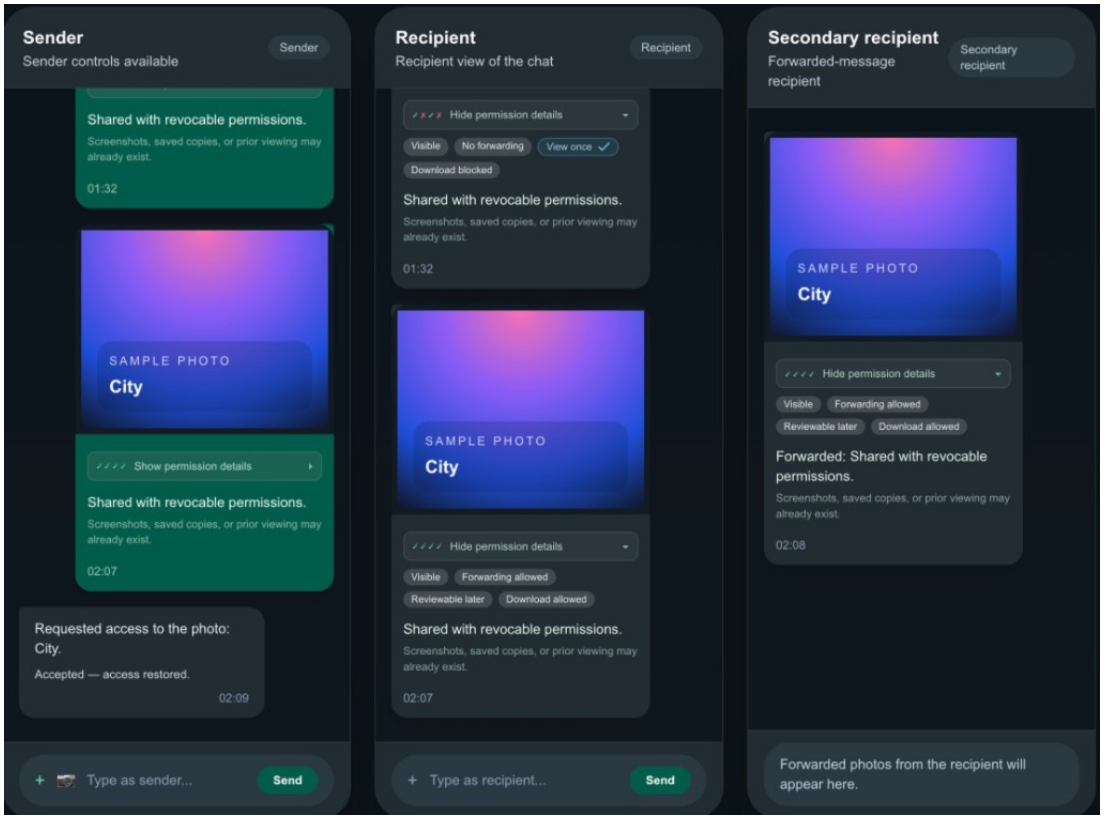
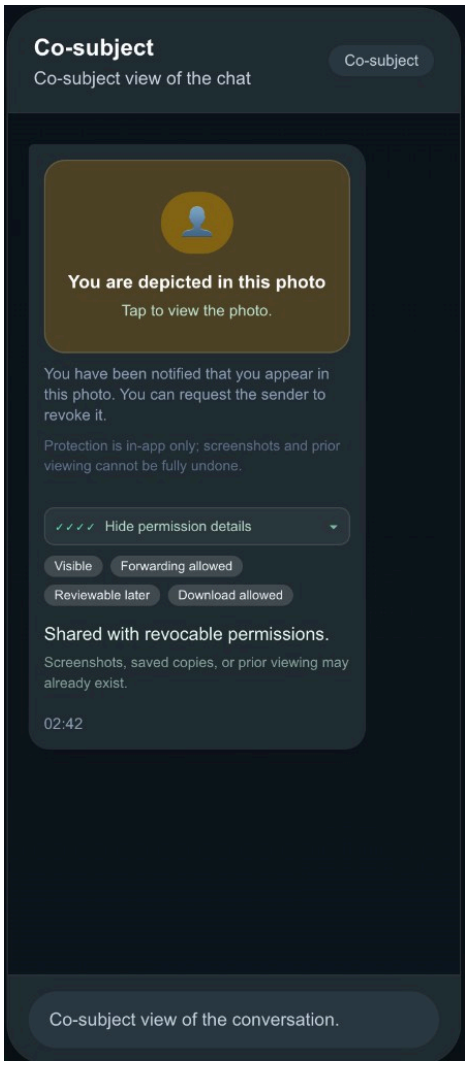


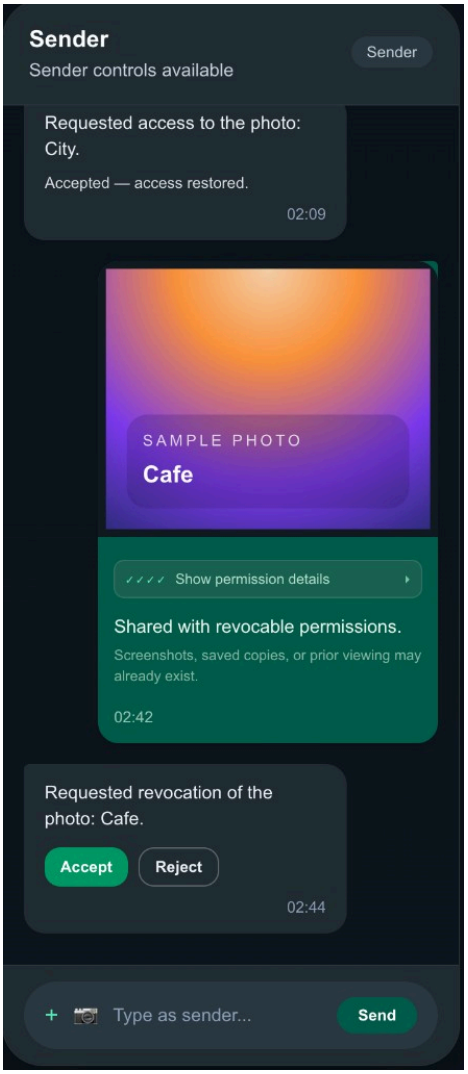
Figure 5.15: Three-view state after the sender accepts the access request. The photo is visible again across all participant views, with permission chips reflecting the restored state.

Co-Subject Notification and Revocation Request

Figure 5.16a shows the appearance of the co-subject notification specified in Section 4.3.5. The co-subject's dedicated handset view renders the notification as a prominent card component instead of an ordinary message bubble which visually distinguishes it from regular chat content so that the depicted person cannot overlook it while scrolling. The card reuses the same permission chip row used across all other stakeholder views so that the co-subject can see the live permission state. Beneath the chips, a limitation cue is displayed using the same persistent warning pattern used in the pre-send composer which tries to reinforce the consistent expectation management across all stakeholder views.



(a) Co-subject notification with depicted-person card and permission chips.



(b) Sender's view after receiving the co-subject's revocation request

Figure 5.16: Co-subject flow: notification (a) and revocation request (b).

If the co-subject selects *Request revocation*, from the photo actions menu the `RequestRevocationPanel` component opens inline in the handset footer. The panel identifies the photo by its caption label and provides an optional message field, following the same request pattern used in the recipient's access-request flow to maintain the consistency across stakeholder roles as a user might be any stakeholder in the flow depending on the context. On the sender's side (Figure 5.16), the request appears as a text message with *Accept* and *Reject* buttons. If the sender accepts, the canonical record is updated into a revoked state and the change inherents to all participant views through the derivation mechanism in Section 5.4 including any forwarded copies that share the same `sourcePhotoId` This cross-context propagation is the same mechanism used for sender-initiated revocation which means co-subject triggered revocations requires no additional state logic beyond routing the request to the

existing `updateSelectedPhotoPermissions` handler.

Also a deployed version can be accessed at `https://smart-photos.vercel.app` (works better full screen).

5.6 Software Testing

Manual Cross-View Verification

After each significant modification to the prototype, it is manually tested in the browser using all four views of stakeholders (co-subject, forwarded recipient, recipient and sender) for cross-view consistency. The exercise verified that a revocation created by the sender is immediately visible in the views of the recipient and forwarded recipient, and that the permission chips are updated appropriately. Manual testing was especially crucial for the visual and layout aspects of the system such as color coded status indicators, animations and responsive scaling since they could not be completely captured through automated testing in a simulated DOM environment. All modifications were documented with version control with git during both phases of the development process, so there was an audit trail of all changes to help identify where regressions occurred.

Automated Unit and Component Tests

To supplement the manual checks described above, an automated test suite was developed and organised into three layers within a dedicated `tests/` directory.

Unit tests verify pure helper functions and the permission model in isolation, confirming for example that `computeEffectiveStatus` treats a past expiry timestamp as revoked regardless of the stored status field and that `getEffectivePermissions` falls back to a message's own permission record when the canonical source is absent from the main thread.

Component tests render individual UI elements such as the photo bubble, composer, picker, time-limit modal, request panels, and text message bubble within a lightweight harness that supplies controlled props. These tests exercise behavioural branches including a revoked photo displaying the "Access changed" placeholder, a view-once photo transitioning to "Opened once" after consumption, the download button appearing disabled when `canDownload` is false, and the co-subject card reflecting both active and revoked states.

Integration-style tests render the full `PhotoRevocationDemo` orchestrator and simulate user journeys across several handset panels, including revoking a photo and observing the recipient subtitle change, forwarding a photo and verifying its appearance on the secondary handset, accepting or rejecting access

and revocation requests, and consuming a view-once photo through the full modal lifecycle.

Edge cases are tested alongside the primary flows, including restoring access on an already-expired time limit (which must also clear the `expiresAt` field), sending an empty text message, cancelling the time-limit modal without applying an expiry, and verifying that permission-detail chips display the correct label variant depending on whether the underlying policy was view-once. At the time of writing the suite comprises all test cases across 16 test files. Table 5.2 summarises the coverage collected by Vitest coverage.

Metric	**Coverage (%)**
Statements	96.22
Branches	90.84
Functions	91.66
Lines	96.22

Table 5.2: Aggregate test coverage across all source files

Scope and Limitations of Testing

The automated suite operates within jsdom, which reproduces DOM structure but does not execute real CSS rendering or browser navigation. No end-to-end framework such as Playwright was introduced, as the additional cost was not justified for a single-page prototype where all views coexist in one render tree. Similarly, because the prototype has no backend, there are no API or database tests. These boundaries are consistent with the project's scope as an interactive prototype. As Myers notes, testing can reveal the presence of defects but never prove their complete absence [41]. Automated tests serve as a regression safety net for the permission logic and UI behaviour, while the broader question of whether users understand and trust that behaviour is addressed through the qualitative evaluation in Chapter 6. Broader prototype limitations and production considerations are discussed in Section 7.2.

Chapter 6

Results/Evaluation

6.1 Brief

This chapter details the qualitative, interview-based evaluation of the Smart Photos prototype. It outlines the methodology, including participant sampling and reflexive thematic analysis and presents five key themes that capture how users interpreted the system.

6.2 Interview-based Evaluation

6.2.1 Methodological Approach

A qualitative interview-based evaluation was conducted using semi-structured interviews. This method was appropriate because the research aimed not only to gather feedback on the prototype, but also to investigate how participants made sense of new privacy controls and how they felt about the balance between stakeholders. Weiss argues that qualitative interviewing is useful when researchers need more than standardised answers, because it allows the interviewer to obtain participants' fuller accounts through explanation and discussion, and to access "perceptions and reactions" that are only available through their own accounts [63]. The semi-structured format supported this by combining a consistent scenario sequence across all participants with enough flexibility to probe misunderstandings, emotional reactions, and unanticipated concerns as they arose [63]. Follow-up probes were used selectively depending on each participant's responses, consistent with Weiss's view that the interviewer should direct discussion toward topics that matter to the study while judging when a response requires elaboration [63].

6.2.2 Participants and Sampling

The selection of participants were chosen using purposive sampling. This approach was appropriate because the study's goal was not to produce a statistically representative account of the general population. Instead, it seeks feedback from people whose daily experiences qualified them to evaluate a prototype for messaging-based photo sharing. Weiss argues that, when a study is concerned with the experiences of people who share a common characteristic, the researcher needs a sample of people who together can represent the population of concern, and that with small qualitative samples it is often preferable to select respondents purposively in order to capture important variation rather than rely on probability sampling [63].

The study specifically targeted adults aged 18-24, as this group is especially relevant to the context of the prototype and to contemporary messaging-based photo-sharing practices. In the UK, 18-24-year-olds remained the most active online in 2025, while 16-24-year-olds used the widest range of online communication platforms of any age group, indicating particularly high engagement with digital communication services [47, 48]. This is directly relevant to the current study because online communication platforms include services used for messaging and calling, and WhatsApp has been identified as both the most popular app for those activities and the highest-reaching messaging and calls service across age groups [46, 47]. Although some usage data is reported for the broader 16-24 category, the present study focuses on the adult subset of that group for practical and ethical reasons.

Nine participants were recruited in total (8 male, 1 female), all of whom were university students aged between 20 and 24. The sample was not gender-balanced, which is acknowledged as a limitation. However, the study did not aim to compare responses across demographic groups; its focus was on how participants interpreted and evaluated a specific interface design. In qualitative research of this kind, the analytical priority is depth of individual accounts rather than demographic representativeness [8, 36]. Malterud, Siersma and Guassora's concept of information power further supports this position: when the study aim is narrow and the sample is specific to the phenomenon under investigation, fewer and less diverse participants can still yield sufficient analytical insight [36]. Gender was recorded descriptively to contextualise the sample but was not used as an analytic category. All participants reported regular use of WhatsApp or similar messaging platforms, with a minimum of six years of experience, ensuring familiarity with the photo-sharing conventions and interface patterns that the prototype builds upon.

A target of 8-10 participants was set for this study. This number was considered appropriate for a focused qualitative evaluation with a relatively specific participant group and a narrow research aim. Rather than seeking statistical outcomes, the study aimed to identify the recurring patterns in how participants understood the prototype. This is supported by methodological literature suggesting

that smaller interview samples can be sufficient in focused studies. Hennink, Kaiser and Marconi found that code saturation was reached at 9 interviews, while Guest, Bunce and Johnson found that thematic saturation occured within the first twelwe interviws, with basic elements of higher-level themes were already present by six interviews [22, 24]. In this case, data saturation was reached by the ninth participant, suggesting that the sample was sufficient to generate meaningful evaluative insight within the scope of the dissertation.

6.2.3 Interview Procedure

Each interview will be conducted individually using two main materials: a written interview guide and a set of screenshots taken from the prototype. The interview guide structures the session into a consent stage, a warm-up section, five scenarios, and a final wrap-up. This use of a guide is consistent with Weiss's (1994) view that an interview guide should function as a prompt and checklist for the interviewer rather than as a script that is followed mechanically [63]. The visual materials consist of fictional example screenshots illustrating the main design states. The interview guide and the scenario slides are shown in Appendix B.

At the start of each session, participants will be read a standardised consent statement explaining the purpose of the study, the nature of the prototype evaluation, and the fact that the participant is not being tested. They will also be informed that they may skip any question, stop the interview at any time, and choose whether or not to be audio-recorded [14]. If consent is given, the interview will be audio-recorded so that responses can later be transcribed and analysed using the participant's exact wording. Verbatim transcription is important because it preserves participants' exact language for analysis [14].

The interview begins with a short warm-up about participants' current photo-sharing behaviour on WhatsApp. This order follows DeJonckheere and Vaughn's (2019) guidance that interviews should begin with easy, context-setting questions before moving into deeper discussion [14]. Participants are then shown a series of prototype scenarios in a fixed order using screenshots. These screenshots are based on the set scenarios which includes pre-send permissions screen, the post-send permission-change screen, the recipient-side view after access has changed, the forwarding/request-access-again flow, and co-subject related flow. For each screen participants asked to describe what they think happening. This is intentional, because participants' first interpretations are part of the evaluative data, especially in relation to understandability. This reflects DeJonckheere and Vaughn's (2019) recommendation that interview questions should be open-ended, with follow-up questions used to explore responses in more detail [14]. Probes are used selectively rather than mechanically, so the interview remains conversational. This approach is consistent with Weiss (1994), who argues that rigidly covering every point in an interview guide can reduce the quality of qualitative data [63]. The interviews use fictional

example photos rather than participants' own personal images. This keeps the scenarios realistic while avoiding unnecessary disclosure of private content, which is appropriate given the potentially sensitive nature of the topic [14]. At the end of the interview, participants are asked a short set of reflection questions about whether the design would increase their sense of control, affect their trust in the platform, and which elements felt most clear, useful, confusing, or uncomfortable. They are also invited to add anything else that stood out.

6.2.4 Data Analysis Method

Interview recordings were transcribed and analysed using reflexive thematic analysis [7, 8]. This approach was appropriate because the evaluation aimed to examine how participants interpreted and judged the prototype, rather than simply count preferences or identify isolated usability issues. Reflexive thematic analysis treats the researcher as an active participant in knowledge production rather than as a neutral coder, which suited the interpretive aims of this study [8].

The analysis used an inductive coding strategy. Pre-defined codebook method was not used, the codes were developed directly from the data through repeated engagement with the transcripts. Quotes capturing relevant perceptions, reactions, and experiences were first extracted, then organised into categories based on recurring patterns across participants. Code names were generated to label these categories, and each quote was matched to the relevant codes. Through iterative review and refinement, the codes were then grouped into five broader themes. This bottom-up approach is consistent with reflexive thematic analysis, in which codes and themes are understood as actively constructed by the researcher through sustained engagement with the dataset [7, 8].

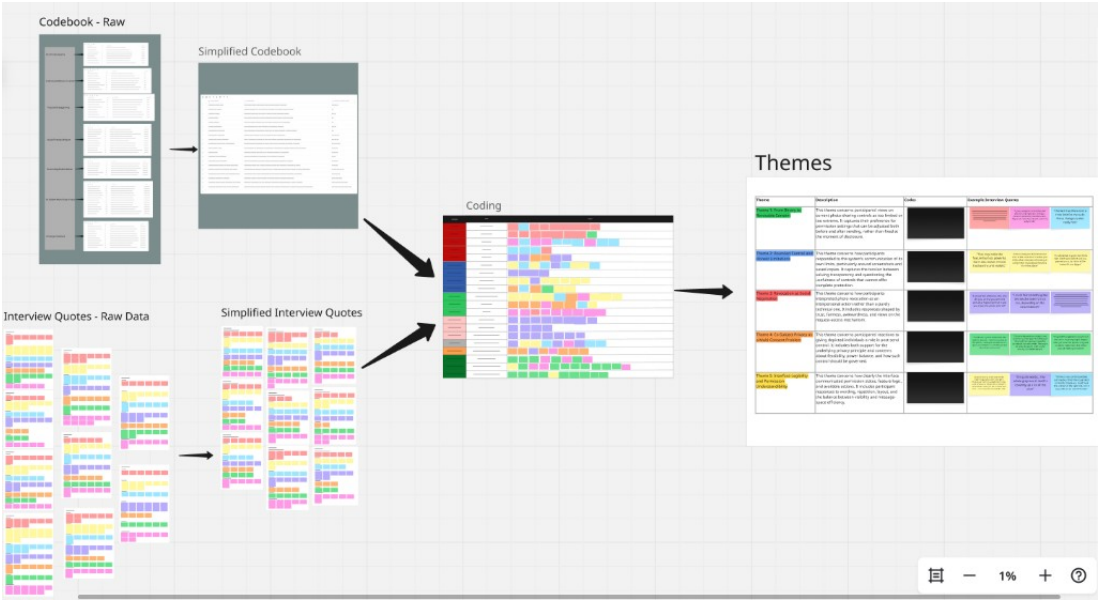


Figure 6.1: Overview of the thematic analysis workflow from raw data to themes (Miro board)

A full view of the analysis board is available here:

Click here to view the Miro board.

Transcripts were first read repeatedly to support familiarisation with the dataset as a whole. Key quotes capturing relevant perceptions, reactions, and experiences were then extracted and placed onto a Miro board as colour-coded sticky notes, with each colour representing the interview section from which the quote originated (warm-up, scenarios 1 through 5, and final reflections). This visual arrangement supported comparison across participants and helped codes and excerpts to be grouped spatially as patterns became more apparent. This process producedapproximately 500 quotes across all nine participants, which were further refined to around 250 analytically relevant quotes.

An initial codebook of 56 fine-grained codes was developed through the inductive process described above. These codes were informed in part by concepts identified in the literature review and research questions, while also being revised and expanded in response to recurring patterns in the interview data. To support efficient and consistent coding across the dataset, the 56 codes were consolidated into 20 working codes grouped into seven conceptual categories: Control and Agency, Understandability and Clarity, Trust and Transparency, Social and Emotional Impact, Forwarding and Redistribution, Co-Subject and Multi-Party Privacy, and UI and Design Feedback. Both the full and simplified codebooks are included in Appendix A.

Coding was conducted primarily at the semantic level, focusing on participants' explicit interpretations and stated reactions, while also attending to broader patterns of meaning where relevant. Each quote was tagged with the relevant code identifiers from the simplified codebook. Codes were then reviewed and grouped into broader candidate themes based on recurring patterns in how participants understood the prototype and evaluated its implications. Candidate themes were checked against the coded data and the wider dataset to ensure they captured meaningful patterns across multiple participants rather than reflecting isolated responses. Themes were refined iteratively until they provided a coherent and internally consistent structure for presenting the evaluation findings. The resulting five themes, their descriptions, contributing codes, and supporting quotes are summarised in Figure 6.2.

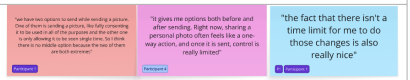
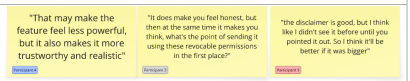
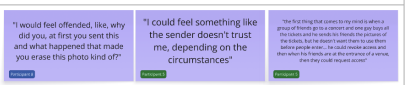
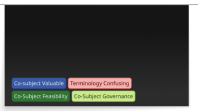
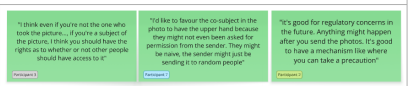
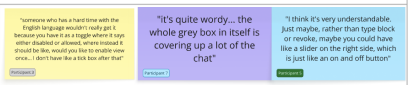
Theme	Description	Codes	Example Interview Quotes
Theme 1: From Binary to Revocable Consent	This theme concerns participants' views on current photo-sharing controls as too limited or too extreme. It captures their preference for permission settings that can be adjusted both before and after sending, rather than fixed at the moment of disclosure.		
Theme 2: Bounded Control and Honest Limitations	This theme concerns how participants responded to the system's communication of its own limits, particularly around screenshots and saved copies. It captures the tension between valuing transparency and questioning the usefulness of controls that cannot offer complete protection.		
Theme 3: Revocation as Social Negotiation	This theme concerns how participants interpreted photo revocation as an interpersonal action rather than a purely technical one. It includes responses shaped by trust, fairness, awkwardness, and views on the request-access mechanism.		
Theme 4: Co-Subject Privacy as a Multi-Consent Problem	This theme concerns participants' reactions to giving depicted individuals a role in post-send control. It includes both support for the underlying privacy principle and concerns about feasibility, power balance, and how such control should be governed.		
Theme 5: Interface Legibility and Permission Understandibility	This theme concerns how clearly the interface communicated permission states, feature logic, and available actions. It includes participant responses to wording, repetition, layout, and the balance between visibility and message-space efficiency.		

Figure 6.2: Summary of the five themes with descriptions, codes, and example quotes.

In line with reflexive thematic analysis, themes were not treated as if they simply emerged from the data, but were actively developed by the researcher through sustained engagement with the dataset and informed by the theoretical context established in Chapter 2 [8]. As a single-researcher project, the analysis did not incorporate formal inter-rater reliability checks. This is consistent with reflexive thematic analysis, which does not treat coding reliability or consensus between coders as markers of quality, but instead positions researcher subjectivity and reflexive engagement with the data as central to the analytic process [8].

6.3 Thematic Analysis Findings

Section analyses and discusses the findings of the thematic analysis in detail.

6.3.1 Theme 1: From Binary to Revocable Consent

The first theme captures how participants percieved current WhatsApp photo sharing controls as inadequate and how the prototype's revisable permissions model strengtens the consent revocation. Participants consistently framed current WhatsApp photo sharing as too binary, typically reduced to either permenant availability or view-once, with no support for changing permissions later. The prototype's ability to configure permissions before and after sending was recognised by all nine participants as an improvement and frequently connected to personal experiences. P1 described this need in ex-

plicitly relational terms: *"when I send some more private pictures to my close friends or my closest relationships... when my status of my relationship changes, I want to go back and undo what I send"* (P1, Warm-up). Furthermore P5 connected this to the risk of accidental disclosure: *"sometimes I can accidentally send pictures to the wrong people and it could have major consequences"* (P5, Warm-up). The shared underlying concern was not simply about deleting content, but about maintaining ongoing agency over shared material as the context evolves.

Several participants explicitly criticised the limited range of existing controls. As P1 put it, *"we have two options... no middle option because the two of them are both extremes"* (P1, Warm-up), while P4 summarised the broader issue by stating that *"in most apps, once the photo is sent, you do not really have much control anymore"* (P4, Warm-up). The prototype addressed this gap by allowing permissions to be configured at the point of sending and revised later.

For some participants, the increased sense of control extended beyond personal comfort to behavioural change. P5 suggested the feature would expand their use of the platform: *"This would definitely increase my use of this platform for photos. It could also make me consider using it more for business, maybe contracts"* (P5, Scenario 2). P2 similarly framed the co-subjectfeature in precautionary and regulatory terms, suggesting it could be useful in legal contexts because it provided a mechanism for showing that permission-related steps had been taken and for reducing future disputes over shared personal data. This indicates that revisable permissions may not only reduce anxiety for existing sharing behaviour but also enable new categories of photo sharing that users currently avoid.

These findings are consistent with the literature on dynamic and ongoing consent reviewed in Chapter2, which argues static authorisation models are poorly suited to interpersonal messaging because the meaning of permission can shift after disclosure [26, 31, 71].

6.3.2 Theme 2: Bounded Honesty and System Limitations

This theme evaluates whether the prototype achieved the design goal set out in Section 4.2.2: communicating revocation as limited but meaningful control rather than implying complete erasure. In practice, the results only partially supported this expectation. For some participants, the screenshot warning and related explanations made the system feel more realistic and therefore more trustworthy, for others, the same honesty weakened the feature's perceived value by foregrounding how little control can be guaranteed after disclosure.

The most positive framing came from P4, who described the warning as making the design feel more credible: *"That may make the feature feel less powerful, but it also makes it more trustworthy and realistic"* (P4, Scenario 1). P7 similarly treated the disclaimer as a useful prompt for reflection before sending, although both P7 and P9 felt, it needed greater visual prominence to function effectively.

This shows that the prototype generally succeeded in managing expectations for participants who had previously accepted revocation as prospective and partial, but that this was strongly dependent on the warning being properly placed and clearly visible.

At the other end of the spectrum, P3 interpreted the same information as undermining the feature's purpose: *"it does make you feel honest, but then at the same time it makes you think, what's the point of sending it using these revocable permissions in the first place?"* (P3, Scenario 1). This scepticism extended to P3's overall willingness to use the feature for sensitive content. P9 did not notice the warning at all until it was pointed out: *"... the disclaimer is good but I didn't see it before until you pointed it out. So I think it'll be better if it was bigger"* (P9, Scenario 1).

The split in responses maps onto a tension identified in the design rationale (Section 4.2.2), where expectation management was positioned as a core design responsibility. The findings suggest that this principle is endorsed by participants who already hold a realistic mental model of digital persistence, but may discourage those who expect stronger guarantees. This is consistent with Schnitzler et al.'s observation that users do not always accurately understand what deletion or revocation actions achieve, and that clearer interface explanations can improve comprehension [53]. The practical implication is that the warning's value depends not only on its content but on its visual prominence and framing. Several participants suggested it should be larger, more prominently placed, or accompanied by icons to ensure it is noticed before sending.

Platform trust responses further illuminate this theme. Participants who accepted the bounded nature of control tended to report increased trust in the platform (P1, P2, P4, P6, P8), reasoning that the feature demonstrated the platform's commitment to user privacy. P1 stated: *"it would make me trust the platform more, because it genuinely shows they want me to have control over my own privacy, which is something you cannot see everywhere in today's world"* (P1, Final Reflections). In contrast, participants who questioned the feature's effectiveness reported no change in platform trust (P3, P5, P7, P9) or expressed broader scepticism about platform intentions (P9): *"it wouldn't change my trust in the platform, because most apps take sensitive information. I still don't trust them completely"* (P9, Final Reflections). P3 reframed the issue as interpersonal rather than institutional: *"the platform, no difference at all. It just makes a difference to the senders and how they interact"* (P3, Final Reflections).

Beyond the screenshot warning itself, five participants identified a deeper technological limitation that the prototype cannot address. Because content is ultimately displayed on a physical screen, it can always be captured using a separate device. As P7 put it, *the photo is still displayed on the device. So you can always take another device and take a picture of the screen"* (P7, Warm-up). Some participants also expected the design to push further against these limits through available countermeasures. P5 suggested that screenshots could render as a black screen, reporting having

seen this implemented elsewhere (P5, Scenario 4). These observations are consistent with Heath et al.'s forensic analysis of ephemeral messaging, which documents multiple preservation routes including forwarding, screenshotting, and photographing content before it disappears [23].

6.3.3 Theme 3: Revocation as a Social Act

Participants did not treat revocation as a neutral technical action. Instead, they seen it as a socially meaningful act that could signal trust-mistrust, changing intentions or shifts in the relationship. The feedbacks ranged from acceptance to offence, with majority of participants suggesting that the social meaning of revocation depended on context, specifically the type of photo and the relationship involved.

Several participants expressed their acceptance by reasoning through reciprocity. So they stated that if they would want the same power as a sender, they should accept it as a recipient. P7 articulated this most clearly: *"there's no harsh feelings,... I'm totally happy with how it works... you don't play the role of just sender or just recipient. When you're using an app like this, you're actively being both"* (P7, Scenario 3).

However, other participants attached more negative meanings to the same action. P2 anticipated feeling *"a bit suspicious"* and *"annoyed"* (P2, Scenario 3), while P5 associated revocation with possible mistrust. The strongest reaction came from P8, who said: *"I would feel offended, like,... at first you sent this and what happened that made you erase this photo?"* (P8, Scenario 3). These responses suggest that even when the interface uses neutral wording, revocation still carries an interpersonal signal that cannot be fully neutralised through design alone.

The request-access feature was designed to balane sender control with recipient-side agency. Again the responses were mixed. P5 reimagined the feature as a timed-access mechanism through a concrete scenario: *"one guy buys all the tickets... he could revoke access and then when friends are at the entrance, they could request access"* (P5, Scenario 3). However, P8 questioned the logic of requesting back something that was deliberately withdrawn: *"it makes things more weird. He revoked it for a reason. Why should I want it again?"* (P8, Scenario 3). P9 raised a practical concern about abuse: *"if the person keeps spamming the request access option, it might be bothersome. So a timer, so they can't request as frequently"* (P9, Scenario 3).

These findings connect to several theoretical perspectives from Chapter 2. The range of emotional reactions is consistent with CPM theory's concept of boundary turbulence, where changes to previously established privacy rules can produce conflict or discomfort [49]. The social meaning participants attached to revocation also supports the design rationale's third principle (Section 4.2.3), which aimed to minimise social friction through neutral wording (a goal that the wording succesfully achieved, as 88% of the participants described it as neutral), but which does not completely eliminate the relational

signal that exists there in the revocation process itself.

6.3.4 Theme 4: Co-Subject Privacy as a Multi-Consent Problem

This theme captures the participants' reactions to the co-subject feature which is the most complex and contradictory set of responses in the evaluation. This feature was simultaneously the most confusing and for several participants the most important feature in the prototype. At first each participant (9/9) couldn't understand the 'co-subject in photo' label and required explanation before they could engage with the concept. However, after the explanation, majority recognised the value of protecting the privacy of depicted individuals.

The terminological confusion was universal and immediate. P1 asked directly: *"co-subject in photo. Can you please explain what is co-subject in photo?"* (P1, Scenario 1). P9 attempted to interpret it through an existing mental model: *"the co-subject is a bit hard to understand, but I think it's like how tagging works on Instagram"* (P9, Scenario 1). The tagging comparison was made independently by four participants (P1, P8, P9, and implicitly P5), suggesting that tagging is the closest available reference point for users encountering multi-party photo permissions for the first time.

Furthermore, participants generally accepted the privacy concern the feature was trying to address. P3 expressed this most strongly: *"even if you're not the one who took the picture, if you're a subject, you should have the rights as to whether or not other people should have access"* (P3, Scenario 5). P9 similarly linked the feature to protection against harm, arguing that equal power between sender and co-subject would be preferable in cases where the depicted person might otherwise be exposed without consent. These responses suggest that participants saw depicted individuals as having a legitimate stake in downstream photo sharing, even when they were not the original sender.

At the same time, participants raised substantial concerns about feasibility. P8 questioned the fundamental premise: *"why would the sender at the first point bother to tag the co-subject?"* (P8, Scenario 5), highlighting that the feature depends entirely on voluntary sender action. P2 suggested automated identification as an alternative, but P5 immediately identified the accuracy risk: *"you could be on the street and another guy looked at the camera and if the app detects that as a co-subject..."* (P5, Scenario 5). This tension between manual tagging (which depends on sender willingness) and automated detection (which introduces accuracy and privacy risks) reflects a challenge also identified by Xu et al., who propose privacy-preserving facial recognition for co-owner detection but acknowledge the dependency on reliable identification [72]. P1 identified a further privacy tension where notifying a co-subject could reveal that the sender had shared their image in a private conversation, potentially harming the privacy of that conversation itself. To reduce friction, P5 proposed persistent consent preferences: *"always accept from this person, or never accept"* (P5, Scenario 5) which would function

as contact-level trust settings, reducing per-photo overhead. Together, these concerns highlights while the co-subject concept is valued in principle, its implementation must navigate a trade-off between identification accuracy, user friction, and sender willingness to participate.

On the question of who should hold the strongest control, opinions differed into three different clusters. The majority (P1, P2, P4, P5, P9) favoured equal or balanced control between the sender and co-subject. P7 and P8 favoured giving the co-subject stronger control, with P7 arguing: *"I'd favour the co-subject to have the upper hand because they might not even been asked for permission. The sender might just be sending it to random people"* (P7, Scenario 5). A third group (P3, P5, P6) advocated a veto model in which any single objection would prevent sharing: P6 stated that *"if there's 5 or 6 people and only one says no, then it can't be sent..."* (P6, Scenario 5).

These findings are consistent with the multi-party privacy conflict literature, which argues that co-owned content creates situations where the person sharing is not the only person exposed to privacy consequences [11]. The divergence in participants' governance preferences reflects a broader problem identified in the literature where co-owned photos cannot be governed adequately through sender-only control. Existing approaches offer partial support for multi-party privacy, but still leave unresolved how conflicting stakeholder interests should be managed once an image begins to circulate [11, 72, 74]. Therefore, the prototype addresses this by giving the depicted person a role after disclosure through a co-subject revocation request, rather than leaving control entirely with the sender. Overall, the findings suggest that co-subject control was conceptually valued, but would require clearer terminology and more carefully engineered governance before it could function in practice.

6.3.5 Theme 5: Interface Legibility and Presentation Efficiency

This theme concerns how permission information was visually presented in the chat interface and how participants evaluated the trade-off between visibility and usability. Unlike earlier themes, which focused on the meaning and social implications of revocable sharing, this theme focuses specifically on interface density, readability, and the practical burden of displaying permissions inline within everyday messaging.

The most consistent concern was the space occupied by permission labels beneath photos. P6 noted: *"imagine there are quite a few photos and it's this long list with all these features. It can be annoying to scroll"* (P6, Scenario 2). P7 raised an accessibility dimension: *"it's quite wordy... it might take a while to process for a user with accessibility issues"* (P7, Scenario 3). P8 described the same issue from the sender's perspective, observing that sending multiple photos with individual permission displays would make the chat difficult to navigate (P8, Scenario 2). Participants suggested several alternatives: P6 proposed a long-press contextual menu following existing WhatsApp patterns, P5 suggested on/off

toggle sliders instead of text-based labels, and P1 preferred permissions hidden behind a tap rather than displayed inline. These suggestions converge on a shared preference which indicates permission information should be accessible but not persistently visible.

The concerns are consistent with established usability guidance. Nielsen's heuristics emphasise that interfaces should avoid displaying information that is irrelevant or rarely needed, since every additional unit of information competes with more important content and reduces its visibility [43]. The same guidance also notes that users should not have to wonder whether different words, situations, or actions mean the same thing, which is relevant to participants' confusion over the distinction between view-once and time-limit controls [43].

Two further clarity issues also emerged. P8 and P9 both perceived view-once and time-limit as overlapping features, with P9 stating: *"viewing it once and putting a time limit, I think they are similar. Both accomplishes the same thing"* (P9, Scenario 1). This suggests the distinction between these two controls needs clearer differentiation in the interface. P8 also noted inconsistency in how revocation was communicated, observing that the revoked state used both passive and active voice and *"basically repeats it three times in different places"* (P8, Scenario 3).

At the same time, majority of the participants recognised that the prototype's visual consistency with WhatsApp's existing patterns supported learnability. P5 noted: *"it's integrated to the WhatsApp system and the WhatsApp user interface, it looks like I'm already familiar with this stuff"* (P5, Final Reflections). This suggests that the design principle of consistency with existing interaction patterns (Section 4.2.5) was effective, and that the challenge is not the permission model itself but the density of its always-visible presentation within the chat.

Theme	**Research Questions**	**Research Gaps**
T1: From Binary to Revisable Control	RQ1, RQ2, RQ5	RG1
T2: Bounded Honesty and System Limitations	RQ1, RQ5	RG2
T3: Revocation as a Social Act	RQ4, RQ5	RG3
T4: Co-Subject Privacy as a Multi-Stakeholder Problem	RQ3, RQ4	RG3, RG4
T5: Interface Legibility and Permission Understandability	RQ3, RQ5	RG1, RG4

Table 6.1: Mapping of themes to research questions and research gaps.

6.4 Summary of Findings

Taken together, the five themes indicate that the prototype's core permission model was well understood and valued as increasing user control for shared photos compared to WhatsApp's current model. This positive overall assessment was also reflected in participants' clarity ratings, which ranged from 3 to 5 with a median of 4. However, responses also showed that this control was understood as socially sensitive and dependent on clear presentation. Participants appreciated the ability to revise permissions after sending, particularly in situations involving sensitive content, accidental disclosure, or changing relationships, but they also recognised that revocation could not fully prevent screenshots or external capture. In addition, revocation was interpreted not only as a technical control but as a socially meaningful act, while the co-subject feature highlighted both the importance and complexity of multi-party consent. Several issues identified across the themes were addressed in the second implementation phase, especially those concerning interface clarity and the communication of revocation.

Chapter 7

Discussion

7.1 Brief

This chapter discusses the thesis as a whole, bringing together the design, implementation, and evaluation of Smart Photos to reflect on what the project shows about consent revocation in private messaging. It considers the wider meaning of the findings, their implications, the limitations of the work, and directions for future research.

7.2 Implications and Recommendations for Stakeholders

The findings of this project carry implications beyond the prototype itself. This section outlines what the report suggests for five stakeholder groups.

Users

The evaluation showed that participants who held realistic expectations about digital persistence were more likely to trust the revocation feature than those who expected complete erasure (Theme 2). This suggests that users benefit from understanding revocable permissions as meaningful but bounded. Such permissions serve to withdraw future platform-mediated access rather than to guarantee that all copies have been removed. Platforms that adopt similar features should therefore invest in clear, upfront communication of limits so that users can make informed sharing decisions. Its effectiveness also relied on the prototype design and how the features were presented to the participants.

Platform Designers

1. **Progressive disclosure for permissions.** The strongest usability concern (Theme 5) was that always-visible permission labels consumed excessive chat space. Designers should consider contextual menus, collapsible panels, or tap-to-reveal patterns that keep permission information accessible without dominating the conversation view.
2. **Neutral, system-mediated revocation language.** The evaluation found that neutral wording was perceived as non-accusatory by the majority of participants (Theme 3). This validates the use of descriptive, system-like phrasing for revocation notices and should be adopted as a design guideline for any post-send permission change.
3. **Familiar terminology for co-subject features.** The term "co-subject" was universally misunderstood at first encounter; four participants independently compared the concept to Instagram tagging (Theme 4). Designers implementing multi-party photo permissions should use language that maps onto existing user mental models rather than introducing unfamiliar terminology.
4. **Permission propagation across forwarding.** The Smart Photos model's approach of linking forwarded copies to the original permission state was understood by participants and maintained sender control across redistribution. This principle should inform how real platforms handle permission inheritance for forwarded media.

Platform Providers

The evaluation findings demonstrates clear user demand for post and pre-send control that goes beyond the current delete-for-everyone model. WhatsApp's recent Advanced Chat Privacy feature moves in this direction but operates at the chat level rather than the photo level [64]. The Smart Photos model shows what photo-level permission management could look like in practice and the evaluation confirms that users both understood and valued this granularity. Providers and developers considering the adoption of a similar feature should consider the following:

- **Honest Communication of Feature Scope:** The Theme 2 finding showed that some users dismissed controls they perceived as incomplete. New permission features should be accompanied by honest explanations of their scope to maintain user trust.
- **Permission Metadata Infrastructure:** The Smart Photos model requires each photo to carry a permission state. In production this means synchronising per-photo state across devices in real time, introducing storage, network, and computational overhead that must be planned for at scale.

- **Encryption Compatibility:** As discussed in specifications, permission metadata would need to travel through the same encrypted channel as message content to avoid weakening the privacy guarantees users expect.
- **Energy and Sustainability:** A deployed system performing real-time permission lookups and cross-device synchronisation at scale would carry a measurable energy cost that providers should factor into their sustainability planning.

Educators

The evaluation revealed that people's ability to use the revocation feature came down to whether they actually understood how digital persistence works in the real world. Several participants were surprised by the structural limits of platform-level control, such as the impossibility of preventing external-device capture. This suggests that consent education would benefit from concrete, scenario-based demonstrations of what revocation can and cannot achieve, rather than abstract policy explanations. The Smart Photos model, with its explicit permission states and visible limitation warnings, could serve as a pedagogical tool for teaching that digital consent is ongoing, partial, and context-dependent.

Policymakers and Regulators

The GDPR establishes a right to withdraw consent, and the UK Online Safety Act creates obligations for platforms to reduce harm from shared content. However, the evaluation shows that technical compliance alone does not satisfy user expectations. Delete-for-everyone, for instance, operates prospectively and cannot address screenshots, saved copies, or prior viewing [50]. Two policy-relevant findings emerge. First, the partial and prospective nature of revocation (Theme 2, Design Principle 2) implies that regulatory standards should set realistic expectations for what "withdrawal of consent" can mean in messaging contexts. Second, the co-subject findings (Theme 4) suggest that multi-party consent in photo sharing is a policy area that current regulation does not adequately address. The evaluation showed that depicted individuals are widely regarded as having a legitimate stake in how shared photos are managed, yet no existing regulatory framework grants them a formal mechanism to request removal from content shared in private messaging.

7.3 Prototype Limitations

The prototype fulfils the functional and non-functional requirements set out in Chapter 4 and all the requirements have been implemented and verified through both manual checks and an automated test suite. The second implementation round addressed the interface and structural issues identified during the interview-based evaluation, while preserving the core permission model unchanged from the specification.

Prototype scope. Because the prototype operated as a front-end simulation without networking, persistent storage, or real encryption (Section 5.2), the evaluation findings apply to the interaction design and permission model rather than to production-level concerns. Participants interacted with a controlled, immediate simulation, meaning their responses may not fully reflect the experience of using revocable permissions under real network conditions where delays or failures could affect trust and perceived reliability. In a networked system, the canonical permission record would need to be stored server-side and queried by every client that holds a copy, raising questions around latency, offline access, and conflict resolution. A recipient viewing a forwarded photo while temporarily disconnected might see a stale permission state, and the system would need to decide whether to block rendering until the latest state is fetched. Platforms with end-to-end encryption such as WhatsApp would face the additional constraint that permission metadata must travel through the same encrypted channel as message content, since storing permissions in plaintext on the server would undermine the privacy guarantees that users expect. Beyond network-level challenges, the evaluation also confirmed that structural limits such as screenshots and external-device capture shaped some participants' willingness to trust the feature (Theme 2), suggesting that user perceptions of control may shift in deployments that incorporate additional countermeasures.

Co-subject scalability. The prototype assumes that the sender manually flags a photo as depicting another person and that the co-subject is already a contact within the same platform. In practice, automated co-subject detection would require on-device face recognition or similar techniques, raising both accuracy and consent concerns. A false positive could notify someone who is not actually depicted, while a false negative could leave a genuine co-subject uninformed. If the co-subject is not a user of the platform at all, there is no channel through which to deliver the notification. These edge cases would need careful design work before the feature could be offered at scale.

Despite these limitations, the findings apply most directly to WhatsApp-like messaging contexts in which users share photos within private conversations and may later wish to revise the terms of that sharing. The underlying principles of dynamic consent, multi-party awareness, expectation management, and neutral communication of access changes are not specific to WhatsApp and could inform the design of revocable sharing features on other platforms where users exchange personal media.

7.4 Future Directions

The limitations and evaluation findings identified above point to several directions for future research, each grounded in a specific gap that the current project could not address.

Live-prototype evaluation. The present study used screenshot-based interviews, which captured participants' interpretive reasoning but not their real-time interaction behaviour. A follow-up study in

which participants use the running prototype to engage in conversations would strengthen ecological validity and reveal usability issues tied to timing, responsiveness, and sequential decision-making that static materials cannot capture.

Broader and more diverse samples. The current sample was not gender-balanced. Evaluating the Smart Photos model with gender-balanced recruitment, and non-student populations would test whether the findings specifically around the social meaning of revocation (Theme 3) and the acceptability of co-subject governance models (Theme 4), generalise beyond the present demographic.

Co-subject identification and governance. The evaluation surfaced an unresolved tension between manual tagging, which depends on sender willingness, and automated detection, which introduces accuracy and privacy risks. Future work could explore hybrid approaches in which lightweight detection prompts the sender to confirm co-subject presence, reducing friction while preserving sender agency. The governance question about whether the co-subjects should hold equal, superior, or veto-level control also warrants dedicated experimental comparison.

Integration with real platform infrastructure. The production limitations discussed in Section 7.3 highlight several open engineering questions that a front-end simulation cannot resolve. Future work could address these by embedding the permission model within a real messaging architecture, beginning with a server-side permission store that clients query on each render, and measuring how revocation latency, offline caching strategies, and encrypted metadata distribution affect both the reliability and perceived trustworthiness of the feature under real network conditions.

Longitudinal adoption. The evaluation captured first-exposure reactions. Diary studies or longitudinal field trials tracking how users engage with revocable permissions over time would address whether the features retain their perceived value as routine settles, or whether users simplify their behaviour by defaulting to a small subset of permission configurations.

Chapter 8

Legal, Social, Ethical and Professional Issues

8.1 Legal Issues

The legal dimension of this project concerns both the handling of research data during evaluation and the extent to which a consent-revocation interface can realistically support safer photo sharing in messaging environments. In the evaluation, the study used preset example imagery and scenario-based tasks rather than participants' own intimate content, and the prototype itself was implemented as a front-end simulation with in-memory state only, without real user accounts, server-side storage, or a persistent database. This meant the project was designed to minimise the collection and retention of personal data. In that sense, the approach is aligned with the data-minimisation and research-data handling principles associated with the UK GDPR and the Data Protection Act 2018, while avoiding any stronger claim that the prototype has been independently verified as legally compliant in the way a deployed commercial platform would need to be [15, 59].

At the design level, the UK GDPR is relevant not because it was written specifically for consent revocation in private messaging, but because it provides a useful benchmark for what meaningful consent and withdrawal should look like in digital systems. Under the GDPR, consent should be freely given, specific, informed and unambiguous, and withdrawal should be as easy as giving consent [15]. This principle supports the aim of the Smart Photos prototype, which was designed to make permissions clearer before sharing and revocation more accessible after sharing. However, as Politou et al. [50] note, revocation operates prospectively rather than retroactively, meaning that later withdrawal

cannot guarantee complete erasure once data have already been circulated. This is especially relevant in messaging contexts such as WhatsApp-style photo sharing, where screenshots, external captures, or recipient-side saving remain possible even if the platform changes future in-app access. The project also relates broadly to the Online Safety Act 2023, since it explores interface mechanisms intended to reduce harm and improve user control but it should still be understood as an academic prototype rather than a system that itself fulfils real platform duties [60]. Finally, the prototype intentionally adopts familiar WhatsApp-like interaction patterns for evaluative realism. It remains as an original prototype and should not imply any affiliation with Meta.

8.2 Ethical Issues

Researching consent revocation in private photo sharing involves a privacy-sensitive topic, since participants are asked to reflect on changing permissions, interpersonal boundaries, and the potential misuse of shared images in messaging environments. Although the project does not focus on explicitly sexual content, engaging human participants still required careful ethical safeguards to protect their privacy, and wellbeing. In line with the British Psychological Society's guidance for internet-mediated research, the evaluation was designed to minimise discomfort and avoid unnecessary disclosure of personal material [9]. Participation was restricted to adults aged 18–24, informed consent was obtained at the start of each session, participants were told that they had flexibility to withdraw at any time. To reduce ethical risk further, the study did not use participants' own data such as private photos, message histories, or real consent disputes. Instead, it relied on fictional example photos, screenshot-based scenarios, and a front-end prototype designed solely for evaluation. Interview recordings were transcribed for analysis, and participant data were anonymised in the reporting process. This approach is also consistent with value-sensitive design principles, which argue that systems should be developed with attention to human values such as privacy, autonomy, and informed control rather than technical functionality alone [17]. Ethically, the prototype was therefore designed not only to explore stronger user control, but also to avoid overstating what such control can achieve once a photo has already been delivered.

Furthermore, the environmental footprint of this project is minimal by design. The prototype was deployes using Vercel but it operates entirely as a front-end application with in-memory state, and has no persistent server, database or cloud infrastructure. The developement process and testing were conducted on a single personal machine using lightweight tools. If a permission model of this kind was integrated into a production messaging platform like WhatsApp, the additional metadata required for per-photo permission states and real time permission updates would introduce a computational overhead.

8.3 Social Issues

Photo sharing in private messaging is not only a technical exchange between a sender and a recipient, but also a social process shaped by audience, and changing interpersonal boundaries. Petronio and Child [49] argue that privacy is negotiated through ongoing boundary management rather than fixed at a single moment, which is highly relevant to this project because consent may change after a photo has already been shared. The social scope of the problem is also wider than a single sender-recipient pair. Cherubini et al. [11] show that photo sharing can create multi-party privacy conflicts when other depicted people are affected by a disclosure, and this becomes even more complex when a photo is forwarded to secondary recipients. In response, the Smart Photos prototype was designed to make these social relationships more visible by showing forwarding context, propagating permission changes across forwarded copies, and allowing depicted co-subjects to request revocation. These features could support more responsible photo-sharing norms by making the social responsibilities that concern consent revocation more explicit. However, the evaluation also showed that revocation can still be interpreted as mistrust or offence depending on the relationship and context. For this reason, the project should be understood as one contribution toward safer and more accountable communication practices in digital messaging.

8.4 Professional Issues

Throughout this project, the work was guided by the BCS Code of Conduct and Code of Good Practice, particularly its emphasis on acting in the public interest, maintaining professional competence and integrity, fulfilling relevant legal and ethical responsibilities, and contributing responsibly to the profession [4]. These principles are directly relevant to a project concerned with privacy-sensitive photo sharing, since poorly designed consent features could mislead users or overstate the level of protection available. In response, the Smart Photos model was developed as a research prototype grounded in literature, iterative design reasoning, and interview-based evaluation, with findings reported cautiously within the scope of the project. This was especially important because the project deliberately focused on interaction design rather than real platform infrastructure, and therefore does not claim to solve technical issues such as screenshots, external-device capture, encryption compatibility, or cross-device synchronisation.

Professional integrity in this project also required transparency and accountability in both implementation and reporting. The prototype was evaluated through a clear research process, supported by automated testing and explicit discussion of limitations, including the restricted scope of a front-end simulation. Rather than presenting revocation as complete control, the report consistently acknowledges that post-sharing permissions are partial and prospective, and it treats expectation management

as a core design responsibility. This approach aligns with broader computing ethics arguments that professional responsibility involves not only building useful systems, but also communicating their limits honestly and considering their wider social consequences [18].

The project also carries economic and commercial relevance. Clearer and more trustworthy consent features could strengthen user confidence in messaging platforms and improve their appeal in a competitive digital market. For platform providers, more responsible photo-sharing controls may also reduce reputational risk and support closer alignment with growing expectations around privacy and user safety. More broadly, the project contributes to the profession by offering evidence-based design recommendations for platform providers, educators, and policymakers on how these new permission models might be handled more responsibly in messaging environments.

Chapter 9

Conclusion and Future Work

This report investigated how WhatsApp could better support revocable consent for photo sharing in private conversations. The findings show that current platform controls are too limited to reflect the ongoing and changing nature of consent after a photo has been sent. In response, this report designed and evaluated **Smart Photos**, which is a prototype that treats photo sharing as an ongoing permission process. It allows permissions to be set before sending and revised afterwards. The evaluation showed that participants generally found this model supportive for user control and consent revocation when compared to existing WhatsApp features. At the same time the project also showed that consent revocation in messaging can only ever be partial. Screenshots, prior viewing and external-device capture remain structural limits that interface design cannot fully prevent. This means that the revocation should be understood as meaningful control over future platform-mediated access rather than a guarantee of complete erasure after delivery. The co-subject findings further suggest that photo-sharing consent is not always only between sender and recipient, but may also involve the interests of other people depicted in the image.

As discussed in the Section 7.4, future work should validate these findings through live-prototype studies where the project is integrated into both web and mobile apps. This way it will be more usefull for, broader and more diverse samples, and it will help testing within real messaging infrastructure over longer-term use.

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Appendix A

Thematic Analysis Materials

This appendix reproduces the reflexive thematic analysis workbook as exported from the project Miro board.

A.1 Interview extracts board (raw)

First page of the full colour-coded transcript extract board.



A.2 Inductive codebook (initial iteration)

First page of the initial 56-code codebook produced through close reading of the interview transcripts.



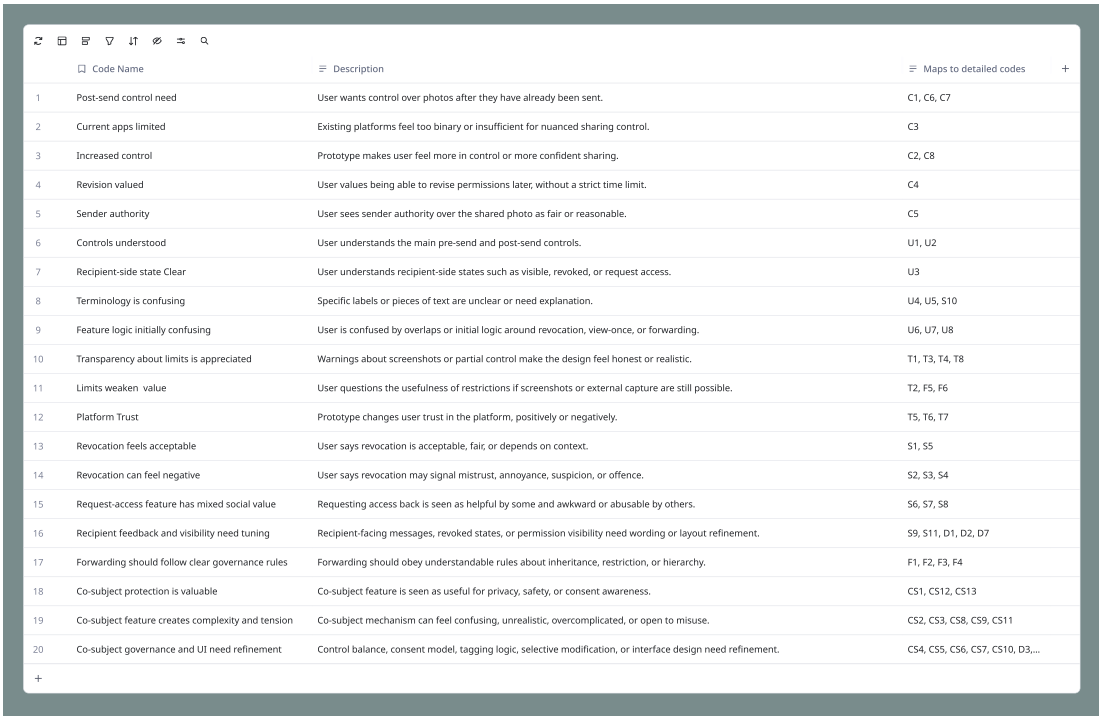
A.3 Interview extracts (refined subset)

First page of the refined extract set used for focused coding.



A.4 Working codebook (simplified)

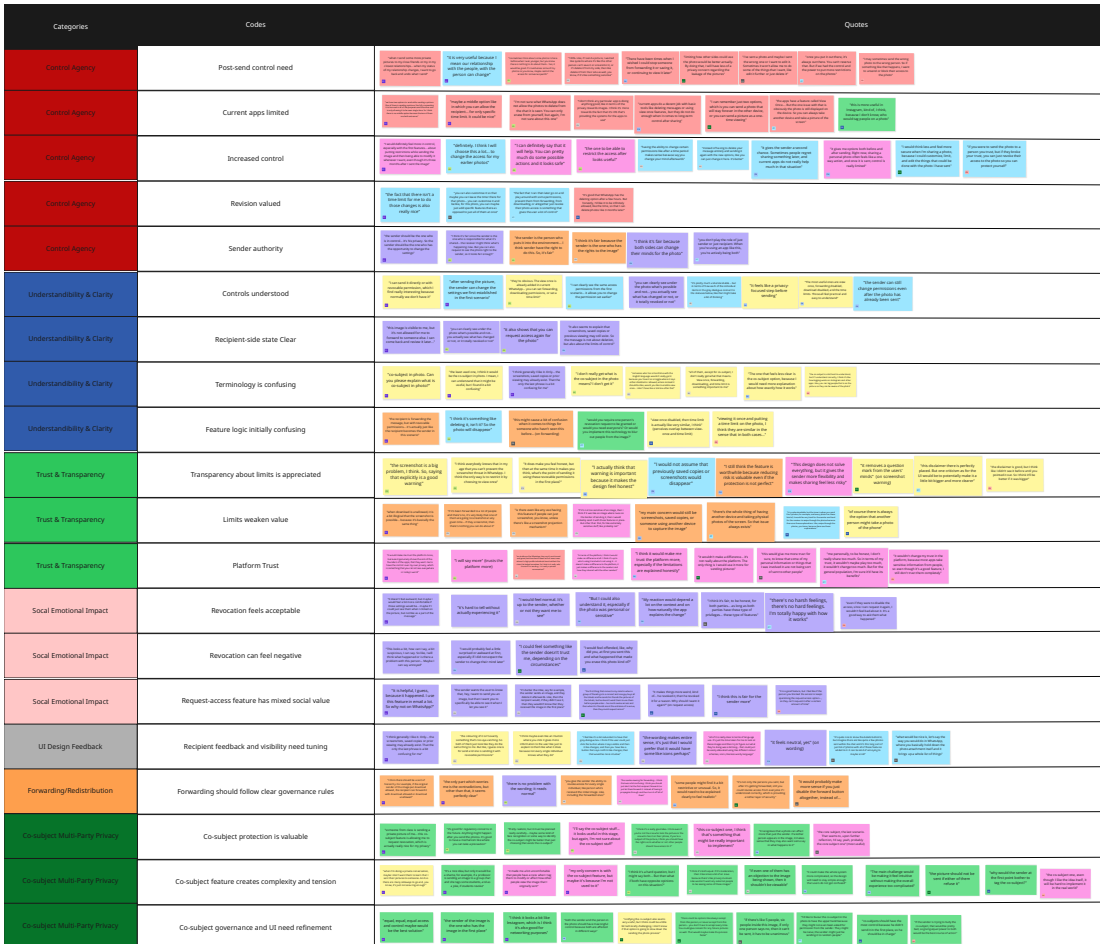
First page of the simplified 20-code working codebook used for cross-case tagging.



	Code Name	Description	Maps to detailed codes
1	Post-send control need	User wants control over photos after they have already been sent.	C1, C6, C7
2	Current apps limited	Existing platforms feel too binary or insufficient for nuanced sharing control.	C3
3	Increased control	Prototype makes user feel more in control or more confident sharing.	C2, C8
4	Revision valued	User values being able to revise permissions later, without a strict time limit.	C4
5	Sender authority	User sees sender authority over the shared photo as fair or reasonable.	C5
6	Controls understood	User understands the main pre-send and post-send controls.	U1, U2
7	Recipient-side state Clear	User understands recipient-side states such as visible, revoked, or request access.	U3
8	Terminology is confusing	Specific labels or pieces of text are unclear or need explanation.	U4, U5, S10
9	Feature logic initially confusing	User is confused by overlaps or initial logic around revocation, view-once, or forwarding.	U6, U7, U8
10	Transparency about limits is appreciated	Warnings about screenshots or partial control make the design feel honest or realistic.	T1, T3, T4, T8
11	Limits weaken value	User questions the usefulness of restrictions if screenshots or external capture are still possible.	T2, F5, F6
12	Platform Trust	Prototype changes user trust in the platform, positively or negatively.	T5, T6, T7
13	Revocation feels acceptable	User says revocation is acceptable, fair, or depends on context.	S1, S5
14	Revocation can feel negative	User says revocation may signal mistrust, annoyance, suspicion, or offence.	S2, S3, S4
15	Request-access feature has mixed social value	Requesting access back is seen as helpful by some and awkward or abusable by others.	S6, S7, S8
16	Recipient feedback and visibility need tuning	Recipient-facing messages, revoked states, or permission visibility need wording or layout refinement.	S9, S11, D1, D2, D7
17	Forwarding should follow clear governance rules	Forwarding should obey understandable rules about inheritance, restriction, or hierarchy.	F1, F2, F3, F4
18	Co-subject protection is valuable	Co-subject feature is seen as useful for privacy, safety, or consent awareness.	CS1, CS12, CS13
19	Co-subject feature creates complexity and tension	Co-subject mechanism can feel confusing, unrealistic, overcomplicated, or open to misuse.	CS2, CS3, CS8, CS9, CS11
20	Co-subject governance and UI need refinement	Control balance, consent model, tagging logic, selective modification, or interface design need refinement.	CS4, CS5, CS6, CS7, CS10, D3,...

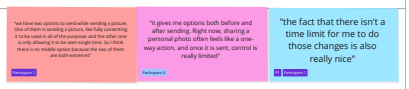
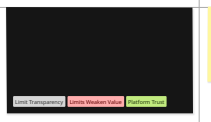
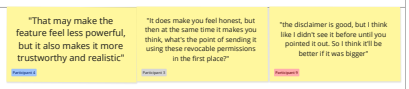
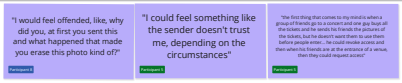
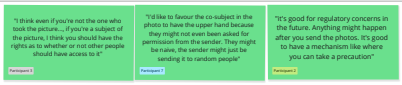
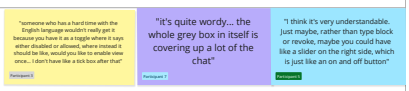
A.5 Quote-to-code coding sheet

First page of the quote-to-code mapping sheet.



A.6 Final themes table

First page of the consolidated themes table (definitions, codes, and example quotations).

Theme	Description	Codes	Example Interview Quotes
Theme 1: From Binary to Revocable Consent	This theme concerns participants' views on current photo-sharing controls as too limited or too extreme. It captures their preference for permission settings that can be adjusted both before and after sending, rather than fixed at the moment of disclosure.		
Theme 2: Bounded Control and Honest Limitations	This theme concerns how participants responded to the system's communication of its own limits, particularly around screenshots and saved copies. It captures the tension between valuing transparency and questioning the usefulness of controls that cannot offer complete protection.		
Theme 3: Revocation as Social Negotiation	This theme concerns how participants interpreted photo revocation as an interpersonal action rather than a purely technical one. It includes responses shaped by trust, fairness, awkwardness, and views on the request-access mechanism.		
Theme 4: Co-Subject Privacy as a Multi-Consent Problem	This theme concerns participants' reactions to giving depicted individuals a role in post-send control. It includes both support for the underlying privacy principle and concerns about feasibility, power balance, and how such control should be governed.		
Theme 5: Interface Legibility and Permission Understandibility	This theme concerns how clearly the interface communicated permission states, feature logic, and available actions. It includes participant responses to wording, repetition, layout, and the balance between visibility and message-space efficiency.		

Appendix B

Interview Guide

This appendix records the scenario-based moderator script used in the evaluation interviews. The wording follows the version spoken in sessions (including the consent statement and scenario prompts). Each numbered question lists the **linked research questions** (Main RQ and RQ1–RQ5) taken from the analysis mapping used alongside the guide: RQ1 addresses current platform limitations; RQ2 expectations and post-sharing concerns; RQ3 clarity of the interface and permission model; RQ4 fairness and social implications; RQ5 trust, safety, control, and realism of revocation. The main research question is stated in Chapter 1.

Research question key

Main RQ

Overall contribution of the proposed design.

RQ1

Limitations of current platforms for controlling shared photos.

RQ2

User expectations, practices, and concerns about post-sharing control.

RQ3

Clarity and understandability of the interface and permission model.

RQ4

Social implications for stakeholders.

RQ5

Trust, safety, perceived control, and realism of revocation.

B.1 Section 0: Mandatory consent statement

Say verbatim:

> "Before we begin, I'd like to confirm your consent to participate in this interview. This interview is part of my research on how messaging platforms might support consent revocation for shared photos. The purpose of this interview is to understand your views and reactions to a design concept. During the interview, I will show you example screens from a prototype. I am not testing you — I am testing the design. There are no right or wrong answers, and honest reactions are very helpful. Some questions are about privacy, control, and what happens to photos after they are shared. You can skip any question or stop the interview at any time. With your permission, I would like to audio-record this interview so I can analyse the responses later. The recording will only be used for this project, and your responses will be anonymised. You can choose to stop the recording or withdraw from the interview at any time. If you're happy to proceed, could you please state your full name and age, and confirm that you agree to be interviewed and recorded?"

B.2 Section 1: Warm-up and current photo-sharing behaviour

1. **Can you tell me a bit about how you usually share photos on apps like WhatsApp?**
   Linked: RQ2.
   Probes optional: private vs group chats; casual vs personal photos; frequency.
2. **When you send a photo to someone, do you usually think about what might happen to it afterwards? If yes, what kinds of concerns or challenges come to mind?**
   Linked: RQ2.
   Probes optional: saving, forwarding, screenshots, long-term retention; active vs incidental thought.
3. **Have you ever wanted more control over a photo after you had already sent it?**
   Linked: RQ1; RQ2.
   Probes optional: type of control wished for; whether the app helped.
4. **What do you think current apps do well or badly when it comes to controlling shared photos?**
   Linked: RQ1.

B.3 Section 2: Scenario 1 — Before sending a photo

Show: photo composer / permission setup (view once, forwarding, download, co-subject, time limit; note on screenshots and saves).

5. **What do you think is happening on this screen?**
   Linked: RQ3.
   Probes optional: purpose of controls; what the sender decides.
6. **What do you think each of these settings means?**
   Linked: RQ3.
   Probes optional: view once; forwarding disabled; download disabled; time limit; co-subject in photo.
7. **Which of these options feel useful to you, and which feel unnecessary or unclear?**
   Linked: RQ2; RQ3.
   Probes optional: realistic pre-send use; effort; missing options.
8. **There is also a warning here saying that screenshots and saves cannot be undone. How do you feel about that?**
   Linked: RQ3; RQ5.
   Probes optional: honesty vs usefulness; whether the warning is wanted.

Moderator note: Only explain controls if the participant is completely stuck; prior interpretation is the primary data.

B.4 Section 3: Scenario 2 — Changing permissions after sending

Show: sent photo with photo menu (revoke/restore, forwarding, download, time limit).

9. **Looking at this screen, what do you think the sender is able to do now?**
   Linked: RQ3.
   Probes optional: what is changeable after send; comparison to deleting a message; revoke vs permission change.
10. **What would you expect to happen if the sender revokes access here?**
    Linked: RQ3; RQ5.
    Probes optional: visibility; recipient notification; offline copies; if already seen.

11. **How understandable does this post-send control feel to you?**

    Linked: RQ3.

    Probes optional: clarity of change; menu affordances; discoverability.

12. **Would this kind of feature make you feel more in control after sending a photo? Why or why not?**

    Linked: RQ5.

    Probes optional: trust; anxiety; realistic use.

Moderator note: Keep focus on clarity, expectations, realism, and perceived control rather than implementation detail.

B.5 Section 4: Scenario 3 — Recipient view, request access, and fairness

Show: recipient-side revoked placeholder (screenshots / prior viewing note).

13. **What do you think this screen is telling the recipient?**

    Linked: RQ3; RQ4.

    Probes optional: clarity of change; sufficiency of explanation.

14. **How would you feel if you were the recipient in this situation?**

    Linked: RQ4; RQ5.

    Probes optional: reasonable vs awkward; fairness tone.

15. **What do you think about the idea that the recipient can request access again?**

    Linked: RQ4.

    Probes optional: comfort as sender or recipient; balance.

16. **Does this feel fair to both people, or more fair to one side than the other?**

    Linked: RQ4.

    Probes optional: sender power; explanation; objection rights.

17. **What do you think about the wording used here?**

    Linked: RQ4.

    Probes optional: neutrality vs blame.

Moderator note: Do not defend the revoked state; negative reactions are informative.

B.6 Section 5: Scenario 4 — Forwarding and residual concerns

Show: forwarding behaviour and secondary recipient where relevant; request-access flow and sender accept/reject if shown.

18. **What do you think this forwarding feature is trying to do?**

    Linked: RQ3.

    Probes optional: blocked vs allowed forwarding; sender rules after forward.

19. **How do you feel about the idea that the sender's rules continue even after the photo is forwarded?**

    Linked: RQ3; RQ5.

    Probes optional: usefulness; realism; over-control.

20. **Even with these features, what concerns would still remain for you?**

    Linked: RQ1; RQ2; RQ5.

    Probes optional: screenshots; prior downloads; residual value of imperfect control.

B.7 Section 6: Scenario 5 — Co-subject request for revocation

Show: co-subject notification and request flow; sender accept/reject.

21. **What do you think about that idea?**

    Linked: RQ4; RQ5.

    Probes optional: fairness; invasiveness; complexity.

22. **For this kind of scenario where the photo includes the sender and other co-subjects, who do you think should have the strongest control in that situation: the sender, the person in the photo, both equally, or someone else?**

    Linked: RQ4.

23. **How realistic does this feel in a real messaging app?**

    Linked: RQ5.

Moderator note: Use only if time allows or if co-subject control is a session focus.

B.8 Section 7: Wrap-up and final reflections

25. **Would this make you feel more in control when sharing personal photos? Why or why not?**

    Linked: RQ5.

26. **Would it make you trust the platform more, less, or no differently?**

    Linked: RQ5.

27. **What part felt most clear or useful?**

    Linked: RQ3; RQ5.

28. **What part felt most confusing, unrealistic, or uncomfortable?**

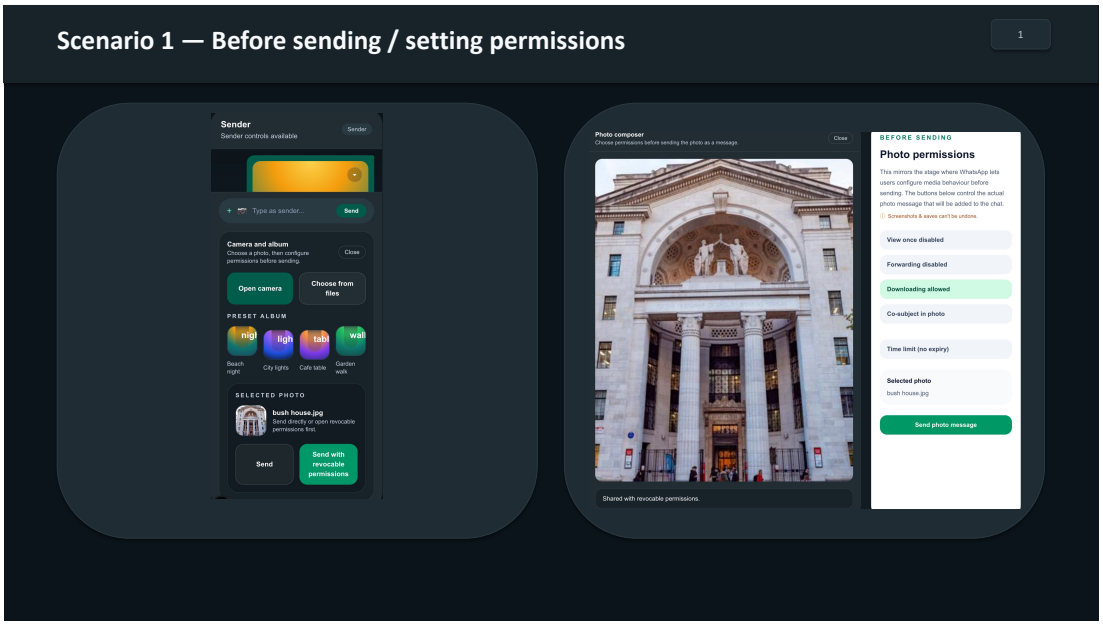
    Linked: RQ1; RQ3; RQ5.

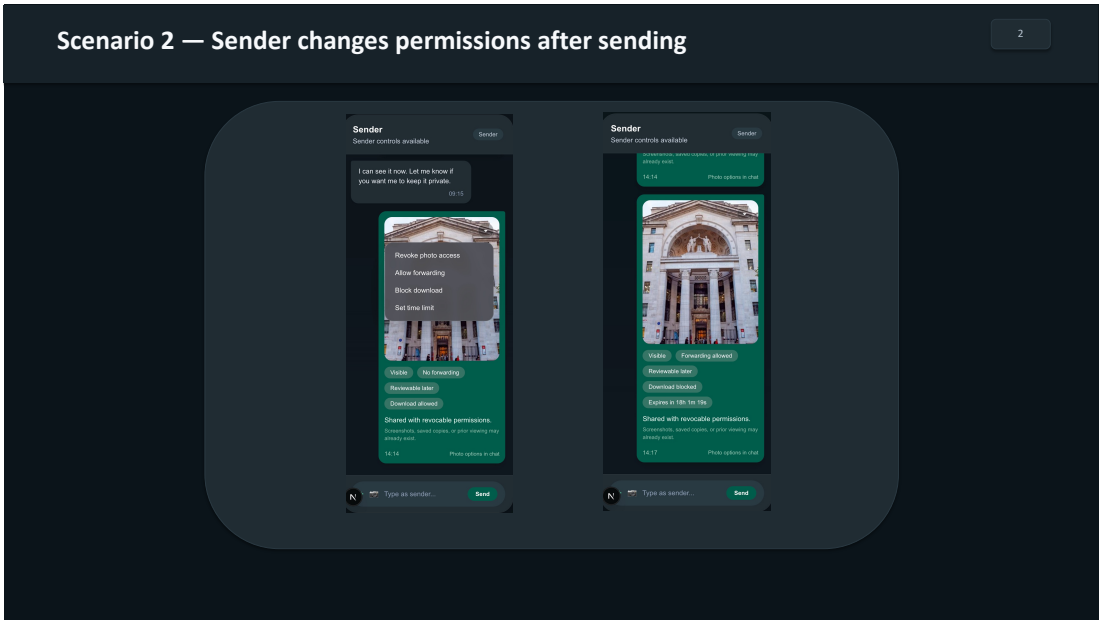
29. **From 1 to 5, how clear do you think everything feels?**

    Linked: RQ3.

B.9 Interview scenario slides (stimulus material)

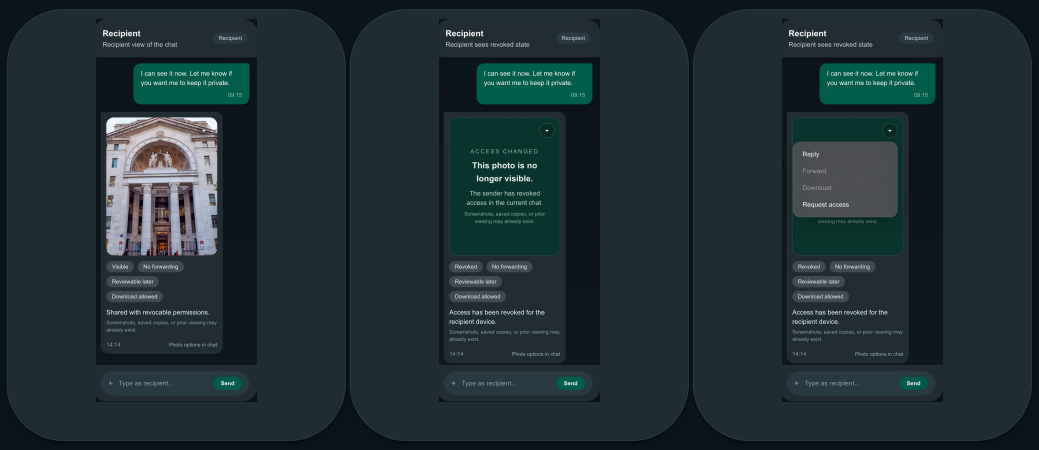
This section reproduces the slide deck *Consent Revocation Interview Scenarios*. The screenshots embedded in these slides were taken from the **first round** of the prototype implementation, before later visual or interaction refinements and they are preserved here to document the materials participants actually saw.





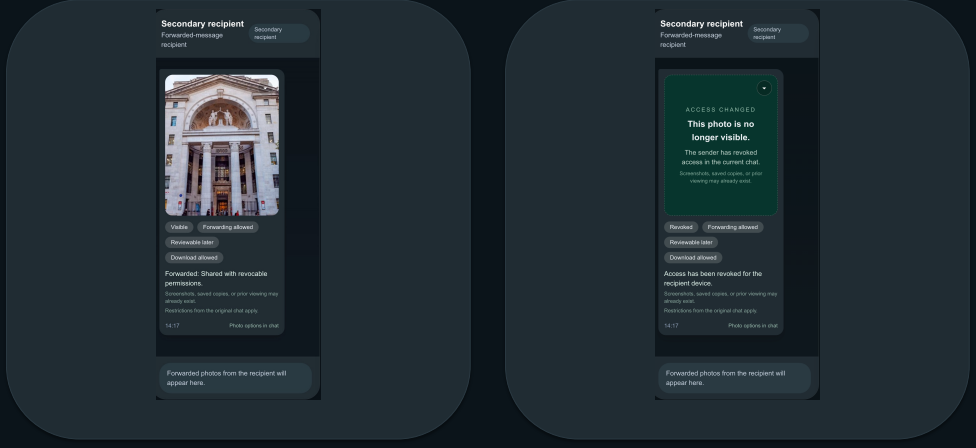
Scenario 3 — Recipient-side impact after access changes

3



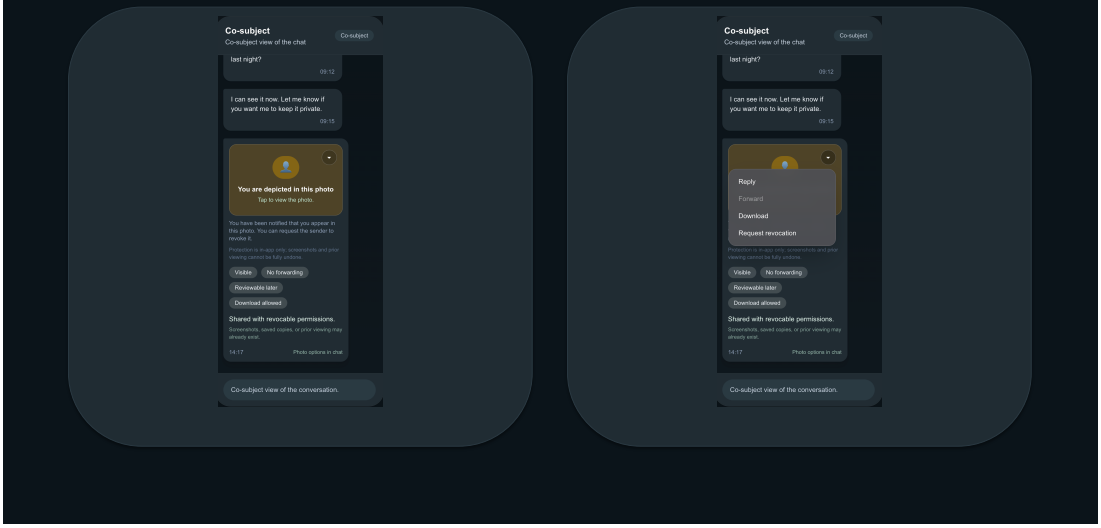
Scenario 4 — Forwarding / secondary recipient

4



Scenario 5 — Co-subject / person depicted in the image

5



Appendix C

User Guide

C.1 Purpose

This appendix explains how to run the **Smart Photos** prototype on a local machine. Also a deployed version can be accessed at `https://smart-photos.vercel.app`. A concise, illustrated tour of every major flow is already given in Section 5.5.

C.2 Prerequisites

- **Node.js** version **20.x** (LTS recommended). The project uses TypeScript tooling aligned with Node 20; older major versions may fail to install or run dependencies.
- **npm** (bundled with Node.js). The instructions below use `npm`

C.3 Install and run

1. Open a terminal and change into the prototype folder:

   ```
   cd photo-revocation-prototype
   ```

2. Install dependencies:

   ```
   npm install
   ```

3. Start the development server:

```
npm run dev
```

4. Open the default address printed in the terminal and make it full screen to have a better view.

C.4 Optional: production build

To run an optimised build (closer to a deployed Next.js app, useful for a stable demo):

```
npm run build
npm run start
```

C.5 Feature checklist (where to see each behaviour)

Feature	**What to do in the running prototype**
Pre-send permissions and screenshot warning	From the sender handset, open the photo picker, choose one or more images, then use *Send with revocable permissions*.
Post-send permission changes / revocation	On the sender handset, open the action menu on a sent photo message and use the post-send options (e.g. revoke access, forwarding, download, time limit). Expand the permission summary row under the bubble if needed.
Recipient view and revoked placeholder	Compare the recipient handset before and after sender-side revocation. Use recipient menu actions such as *Forward*, *Download*, or *Request access* when the design allows them.
View-once consumption	On the recipient handset, open a view-once photo via *Open*, then close the viewer and observe the consumed state on the bubble.
Forwarding and access request	Forward from a recipient handset when permitted. After revocation, use *Request access* and respond on the sender handset with *Accept* or *Reject*. Observe the forwarded-recipient panel for propagated state.
Co-subject flow	Use the co-subject handset panel. Read the notification card, open the photo menu, and use *Request revocation* where available. Complete *Accept*/*Reject* on the sender handset.

C.6 Optional: automated tests

```
npm test:coverage
```

Further detail on testing is given in Section 5.6.

Appendix D

Source Code

D.1 Instructions

Complete source code listings must be submitted as an appendix to the report. The project source codes are usually spread out over several files/units. You should try to help the reader to navigate through your source code by providing a "table of contents" (titles of these files/units and one line descriptions). The first page of the program listings folder must contain the following statement certifying the work as your own: "I verify that I am the sole author of the programs contained in this folder, except where explicitly stated to the contrary". Your (typed) signature and the date should follow this statement.

All work on programs must stop once the code is submitted to KEATS. You are required to keep safely several copies of this version of the program and you must use one of these copies in the project examination. Your examiners may ask to see the last-modified dates of your program files, and may ask you to demonstrate that the program files you use in the project examination are identical to the program files you have uploaded to KEATS. Any attempt to demonstrate code that is not included in your submitted source listings is an attempt to cheat; any such attempt will be reported to the KCL Misconduct Committee.

You may find it easier to firstly generate a PDF of your source code using a text editor and then merge it to the end of your report. There are many free tools available that allow you to merge PDF files.